

Practical: - 0

Q] Introduction to Android, Introduction to Android Studio IDE, Application Fundamentals:

Creating a Project, Android Components, Activities, Services, Content Providers, Broadcast Receivers, Interface overview, Creating Android Virtual device, USB debugging mode, Android Application Overview. Simple “Hello World” program.

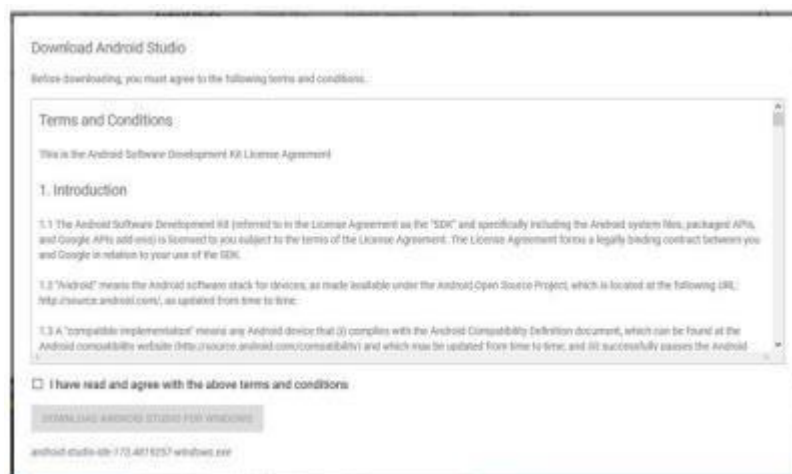
Introduction to Android

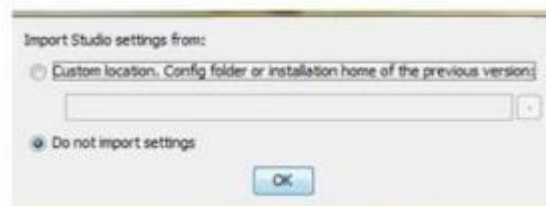
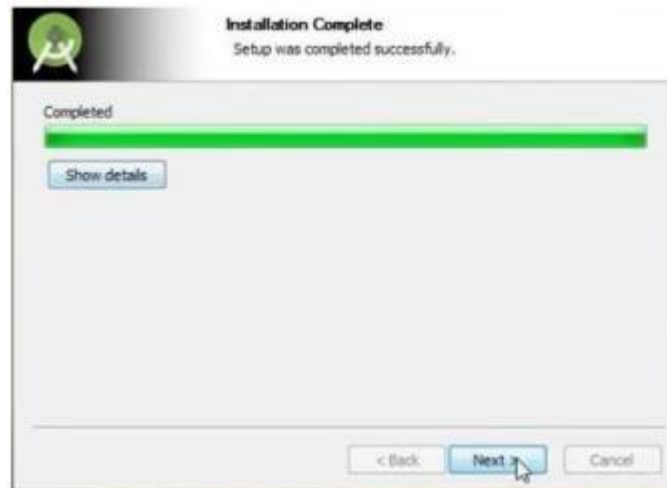
Android is an open-source mobile operating system developed by Google, primarily used for smartphones and tablets. It is built on the Linux kernel and allows developers to create apps using languages like Java and Kotlin. Android supports a wide variety of applications, ranging from simple utilities to complex games, and offers a flexible platform for developers.

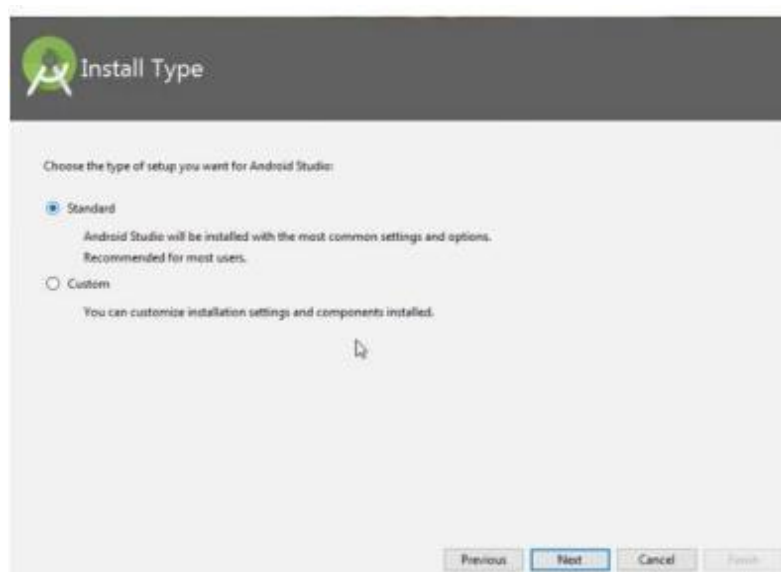
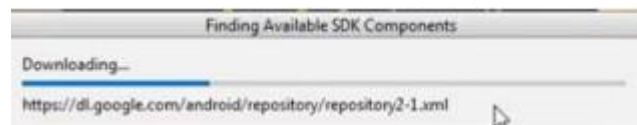
Introduction to Android Studio IDE

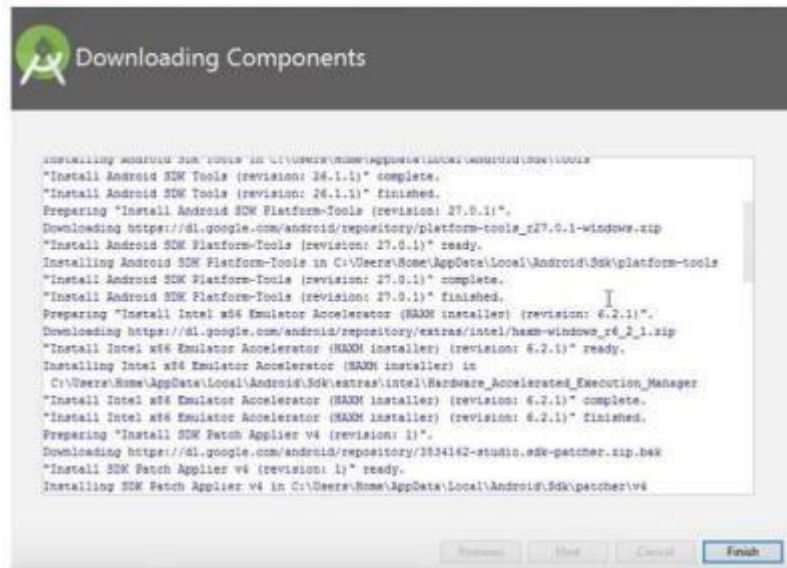
Android Studio is the official IDE for Android app development. Developed by Google, it provides all the tools needed to build, test, and debug Android applications. Android Studio supports both Java and Kotlin, includes a powerful code editor, an emulator for testing apps, a UI designer, and various performance tools to streamline the development process.

Installation

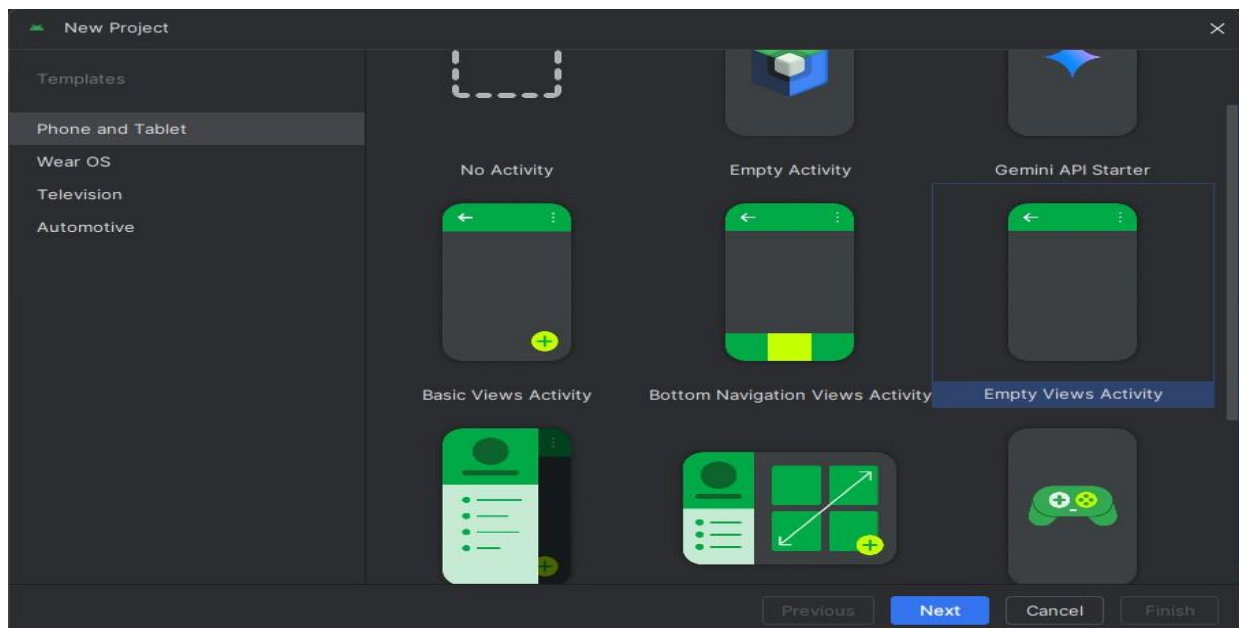
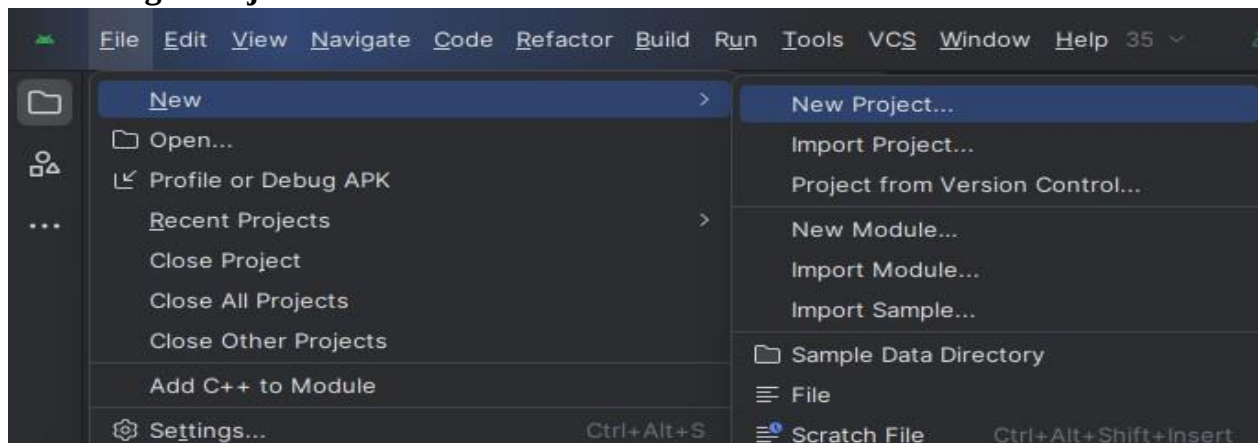


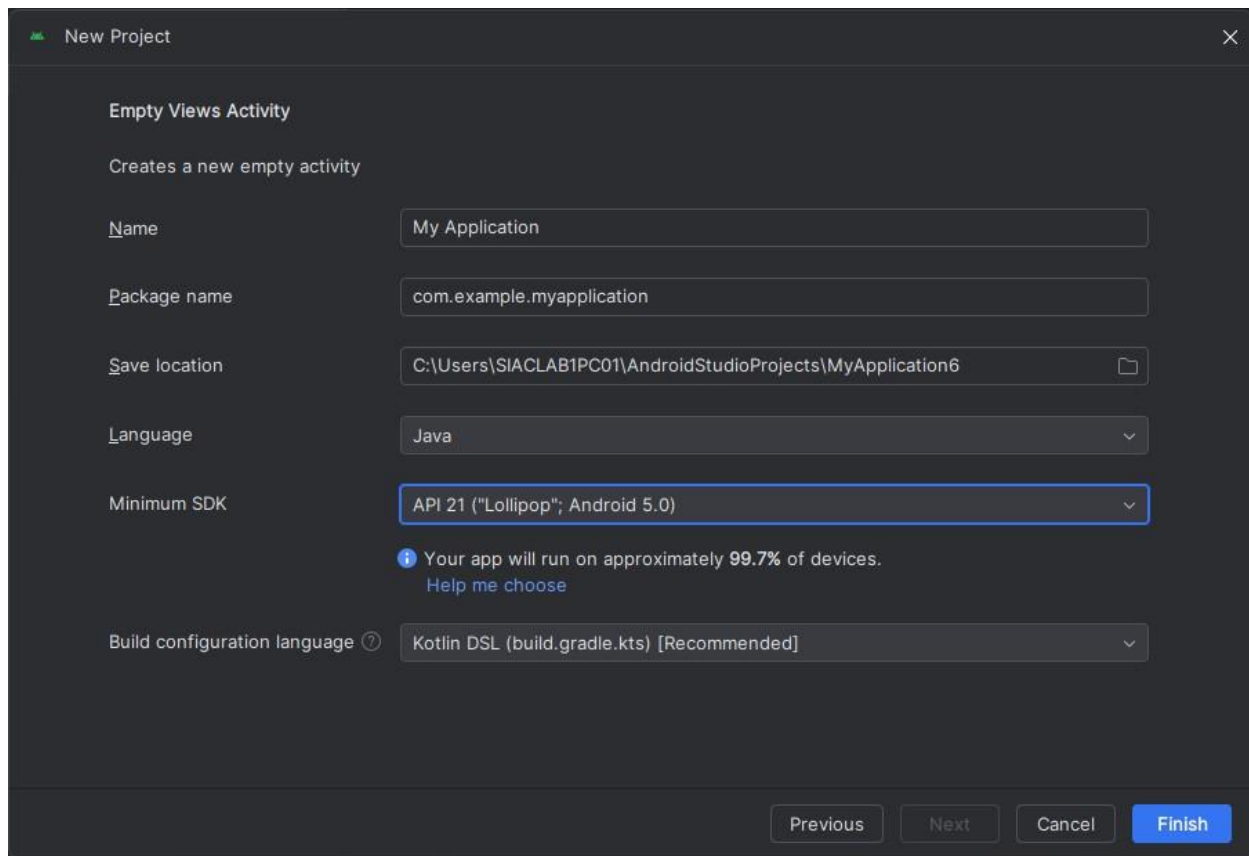






a. Creating a Project

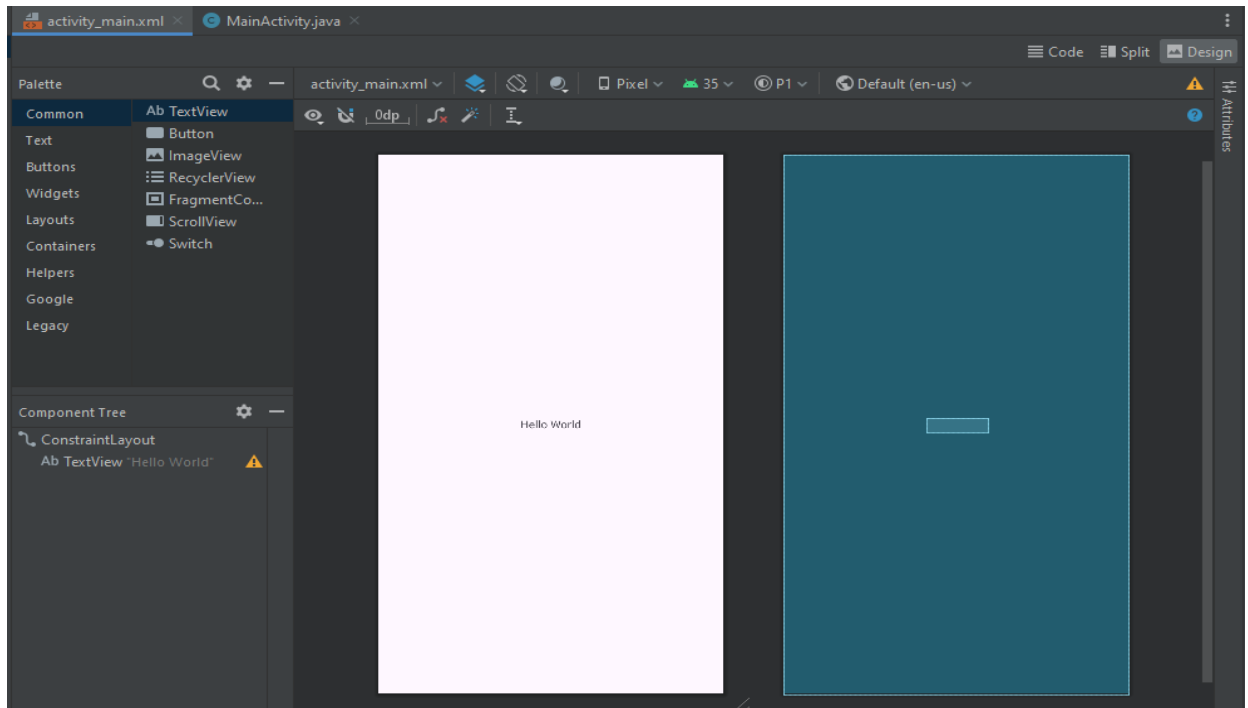




b. Simple “Hello World” program

(activity_main.xml)

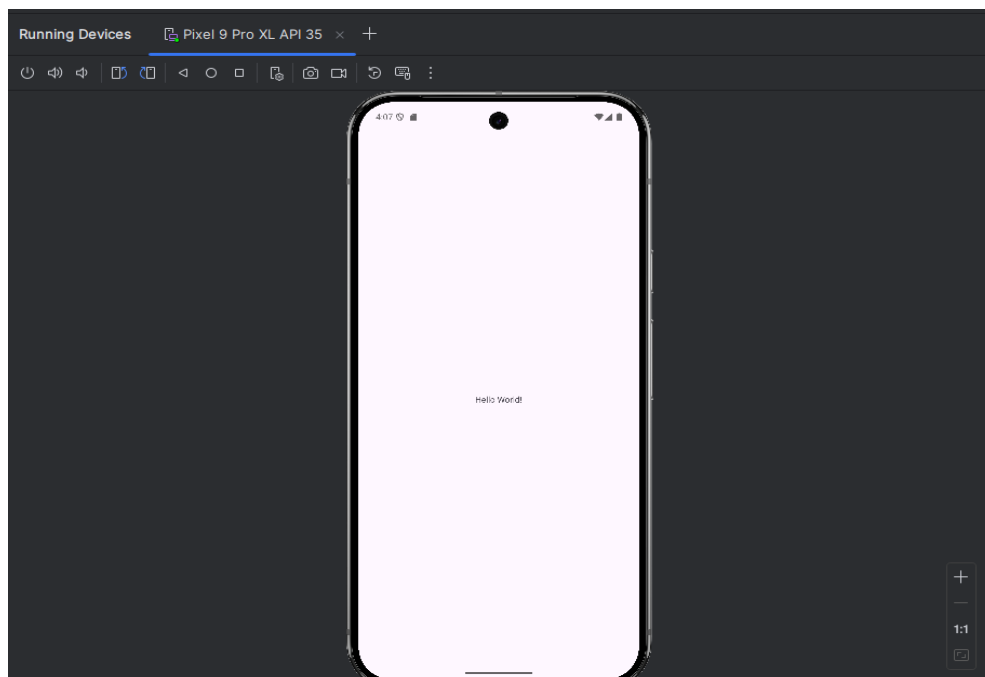
```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Hello World"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintEnd_toEndOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />
</androidx.constraintlayout.widget.ConstraintLayout>
```



(MainActivity.java)

```
package com.example.p1;
import androidx.appcompat.app.AppCompatActivity;
import android.os.Bundle;
public class MainActivity extends AppCompatActivity {
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

OUTPUT



Practical: -1

Q] Programming Resources Android Resources:

(Color, Theme, String, Drawable, Dimension, Image)

a) Defining Color Property

Create a new project and go to: ProjectName>App>src>main>res>values>colors.xml
Android Resources: (Color, Theme, String, Drawable, Dimension, Image)

```
</> colors.xml × © MainActivity.java
1  <resources>
2  <color name="colorPrimary">#6200EE</color>
3  <color name="colorPrimaryDark">#3700B3</color>
4  <color name="colorAccent">#03DAC5</color>
5  <color name="myTextColor">#FF5722</color>
6  </resources>
```

B) Defining string property

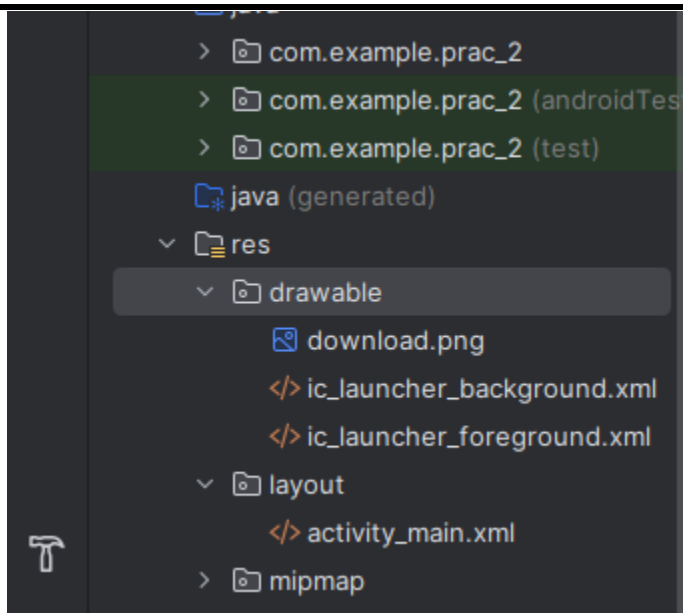
Create a new project and go to: ProjectName>App>src>main>res>values>strings.xml
(Strings.xml)

```
</> strings.xml × © MainActivity.java
1  <resources>
2  <string name="app_name">Resource Demo App</string>
3  <string name="hello_text">Hello from Java Resources!</string>
4  </resources>
```

c) Adding images and dimensions

For adding a new image file, do the following:

ProjectName>App>src>main>res>drawable>Right-click and paste the images



```
</> Dimens.xml x MainActivity.java
1 <resources>
2   <dimen name="text_size_large">24sp</dimen>
3   <dimen name="padding_standard">16dp</dimen>
4 </resources>
```

```
</> v24\rounded_box.xml x MainActivity.java
1
2 <shape xmlns:android="http://schemas.android.com/apk/res/android"
3   android:shape="rectangle">
4   <solid android:color="@color/colorAccent"/>
5   <corners android:radius="12dp"/>
6   <padding android:left="12dp"
7     android:top="12dp"
8     android:right="12dp"
9     android:bottom="12dp"/>
10 </shape>
```

D) Defining Theme Property

Create a new project and go to: ProjectName>App>src>main>res>values>styles.xml (Styles.xml)

</> themes.xml x MainActivity.java

```
1 <resources xmlns:tools="http://schemas.android.com/tools">|
2   <style name="Theme.Practical1"
3     parent="Theme.Material3.DayNight.NoActionBar">
4     <item name="colorPrimary">@color/colorPrimary</item>
5     <item name="colorPrimaryDark">@color/colorPrimaryDark</item>
6     <item name="colorAccent">@color/colorAccent</item>
7   </style>
8 </resources>
```

MainActivity.java

```
package com.example.practical1;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        EdgeToEdge.enable(this);
        setContentView(R.layout.activity_main);

        ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main),
(v, insets) -> {
            Insets systemBars =
insets.getInsets(WindowInsetsCompat.Type.systemBars());
            v.setPadding(systemBars.left, systemBars.top,
                systemBars.right, systemBars.bottom);
            return insets;
        });
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:padding="@dimen/padding_standard">
```

```
tools:context=".MainActivity">

<!-- Title Text -->
<TextView
    android:id="@+id/titleText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Resource Demo App"
    android:textSize="26sp"
    android:textStyle="bold"
    android:textColor="@color/colorPrimary"
    app:layout_constraintTop_toTopOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Image -->
<ImageView
    android:id="@+id/myImage"
    android:layout_width="140dp"
    android:layout_height="140dp"
    android:layout_marginTop="16dp"
    android:src="@drawable/my_image"
    android:contentDescription="@string/app_name"
    app:layout_constraintTop_toBottomOf="@id/titleText"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Main Text -->
<TextView
    android:id="@+id/myText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="16dp"
    android:text="@string/hello_text"
    android:textSize="@dimen/text_size_large"
    android:textColor="@color/myTextColor"
    android:background="@drawable/rounded_box"
    android:padding="@dimen/padding_standard"
    app:layout_constraintTop_toBottomOf="@id/myImage"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintEnd_toEndOf="parent"/>

<!-- Extra Info Text -->
<TextView
    android:id="@+id/infoText"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:layout_marginTop="12dp"
    android:text="***** PRACTICAL_1 *****"
    android:textColor="@color/colorPrimaryDark"
```

```
app:layout_constraintTop_toBottomOf="@id/myText"  
app:layout_constraintStart_toStartOf="parent"  
app:layout_constraintEnd_toEndOf="parent"/>
```

```
<!-- Button 1 -->
```

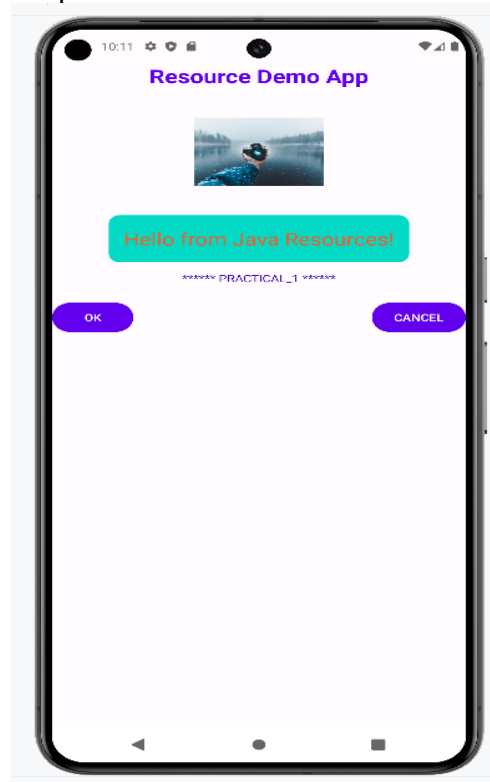
```
<Button  
    android:id="@+id/btnOk"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="OK"  
    android:layout_marginTop="20dp"  
    app:layout_constraintTop_toBottomOf="@id/infoText"  
    app:layout_constraintStart_toStartOf="parent"/>
```

```
<!-- Button 2 -->
```

```
<Button  
    android:id="@+id/btnCancel"  
    android:layout_width="wrap_content"  
    android:layout_height="wrap_content"  
    android:text="CANCEL"  
    android:layout_marginTop="20dp"  
    app:layout_constraintTop_toBottomOf="@id/infoText"  
    app:layout_constraintEnd_toEndOf="parent"/>
```

```
</androidx.constraintlayout.widget.ConstraintLayout>
```

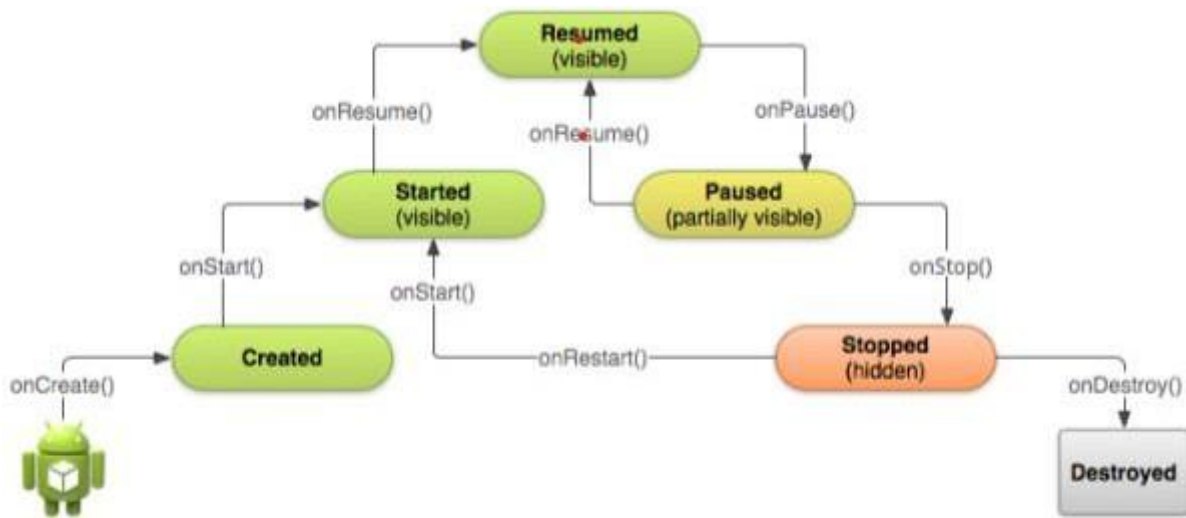
Output:



PRACTICAL-2

Q] Programming Activities and fragments

Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.



1. **onCreate():** Called by the OS when the activity is first created. This is where you initialize any UI elements or data objects. You also have the saved InstanceState of the activity that contains its previously saved state, and you can use it to recreate that state.
2. **onStart():** Just before presenting the user with an activity, this method is called. It's always followed by **onResume()**. In here, you generally should start UI animations, audio based content or anything else that requires the activity's contents to be on screen.
3. **onResume():** As an activity enters the foreground, this method is called. Here you have a good place to restart animations, update UI elements, restart camera previews, resume audio/video playback or initialize any components that you release during **onPause()**
4. **onPause():** This method is called before sliding into the background. Here you should stop any visuals or audio associated with the activity such as UI animations, music playback or the camera. This method is followed by **onResume()** if the activity returns to the foreground or by **onStop()** if it becomes hidden
5. **onStop():** This method is called right after **onPause()**, when the activity is no longer visible to the user, and it's a good place to save data that you want to commit to the disk. It's followed by either **onRestart()**, if this activity is coming back to the foreground, or **onDestroy()** if it's being released from memory.
6. **onRestart():** Called after stopping an activity, but just before starting it again. It's always followed by **onStart()**.
7. **onDestroy():** This is the final callback you'll receive from the OS before the activity is destroyed. You can trigger an activity's destruction by calling **finish()**, or it can be triggered by the system when the system needs to recoup memory. If your activity includes any background threads or other long-running resources, destruction could lead to a memory leak if they're not released, so you need to remember to stop these processes here as well.

Activity Life Cycle, Activity methods, Multiple Activities, Life Cycle of fragments and multiple fragments.

2A) ActivityLifeCycle

MainActivity.java

```
package com.example.practical2;
import android.os.Bundle;
import android.util.Log;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    String TAG = "ActivityLifeCycle";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d(TAG, "onCreate called");
    }

    @Override
    protected void onStart() {
        super.onStart();
        Log.d(TAG, "onStart called");
    }

    @Override
    protected void onResume() {
        super.onResume();
        Log.d(TAG, "onResume called");
    }

    @Override
    protected void onPause() {
        super.onPause();
        Log.d(TAG, "onPause called");
    }

    @Override
    protected void onStop() {
        super.onStop();
    }
}
```

```

        Log.d(TAG, "onStop called");
    }

    @Override
    protected void onRestart() {
        super.onRestart();
        Log.d(TAG, "onRestart called");
    }

    @Override
    protected void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy called");
    }
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Activity Life Cycle Demo"
        android:textSize="20sp" />
</LinearLayout>

```

Output:

2026-02-04 09:56:13.639	8584-8584	ActivityLifeCycle	com.example.practical2	D onCreate called
2026-02-04 09:56:13.660	8584-8584	ActivityLifeCycle	com.example.practical2	D onStart called
2026-02-04 09:56:13.662	8584-8584	ActivityLifeCycle	com.example.practical2	D onResume called

2B) Life Cycle of fragments and multiple fragments

1. Create a new project
2. Add a new Fragment ProjectName>App>src>main>Java
3. Right click on Java and select 'Add Fragment'>Add Fragment(Blank)
4. Give a name to the fragment and click Finish
5. Go to the new file and press Alt + _ Insert and click on

'Generate'>Override Methods (Or directly press Ctrl+O)

6. Select the lifecycle methods that we need to override, which are as follows:

- onAttach(context:Context):void
- onCreate(savedInstanceState:Bundle):void
- onCreateView(view:View savedInstanceState:Bundle):void
- onActivityCreated(savedInstanceState:Bundle):void
- onStart():void
- onResume():void
- onPause():void
- onStop():void
- onDestroyView():void
- onDestroy():void

7. Now go to, LifeOfFragment.java (New Fragment File)

MainActivity.java

```
package com.example.practical2;
import android.os.Bundle;
import android.util.Log;
import android.widget.Button;
import androidx.appcompat.app.AppCompatActivity;
import androidx.fragment.app.FragmentManager;
import androidx.fragment.app.FragmentTransaction;

public class MainActivity extends AppCompatActivity {

    String TAG = "MainActivity";

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Log.d(TAG, "onCreate");

        Button btnLoadFragment = findViewById(R.id.btnLoadFragment);

        btnLoadFragment.setOnClickListener(v -> {
            Log.d(TAG, "Button clicked - loading Fragment");

            Fragment1 fragment = new Fragment1();
```

```

        FragmentManager fm = getSupportFragmentManager();
        FragmentTransaction ft = fm.beginTransaction();
        ft.replace(R.id.fragmentContainer, fragment);
        ft.commit();
    });
}

@Override protected void onStart() {
    super.onStart();
    Log.d(TAG, "onStart");
}

@Override protected void onResume() {
    super.onResume();
    Log.d(TAG, "onResume");
}

@Override protected void onPause() {
    super.onPause();
    Log.d(TAG, "onPause");
}

@Override protected void onStop() {
    super.onStop();
    Log.d(TAG, "onStop");
}

@Override protected void onDestroy() {
    super.onDestroy();
    Log.d(TAG, "onDestroy");
}
}

```

Fragment1.java

```

package com.example.practical2;
import android.os.Bundle;
import android.util.Log;
import android.view.LayoutInflater;
import android.view.View;
import android.view.ViewGroup;
import androidx.annotation.NonNull;
import androidx.annotation.Nullable;
import androidx.fragment.app.Fragment;

```

```
public class Fragment1 extends Fragment {

    String TAG = "Fragment1";

    @Override
    public void onCreate(@Nullable Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        Log.d(TAG, "onCreate");
    }

    @Nullable
    @Override
    public View onCreateView(
        @NonNull LayoutInflater inflater,
        @Nullable ViewGroup container,
        @Nullable Bundle savedInstanceState) {

        Log.d(TAG, "onCreateView");
        return inflater.inflate(R.layout.fragment_1, container, false);
    }

    @Override
    public void onStart() {
        super.onStart();
        Log.d(TAG, "onStart");
    }

    @Override
    public void onResume() {
        super.onResume();
        Log.d(TAG, "onResume");
    }

    @Override
    public void onPause() {
        super.onPause();
        Log.d(TAG, "onPause");
    }

    @Override
    public void onStop() {
        super.onStop();
    }
}
```

```

        Log.d(TAG, "onStop");
    }

    @Override
    public void onDestroyView() {
        super.onDestroyView();
        Log.d(TAG, "onDestroyView");
    }

    @Override
    public void onDestroy() {
        super.onDestroy();
        Log.d(TAG, "onDestroy");
    }
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <Button
        android:id="@+id/btnLoadFragment"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Load Fragment" />

    <FrameLayout
        android:id="@+id/fragmentContainer"
        android:layout_width="match_parent"
        android:layout_height="match_parent"
        android:layout_marginTop="16dp" />

</LinearLayout>

```

Fragment_1.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"

```

```
        android:gravity="center"
        android:orientation="vertical">

        <TextView
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="This is Fragment 1"
            android:textSize="20sp" />
    </LinearLayout>
```

Output:

2026-02-04 10:14:06.481	9448-9448	MainActivity	com.example.practical2	D	onCreate
2026-02-04 10:14:06.493	9448-9448	MainActivity	com.example.practical2	D	onStart
2026-02-04 10:14:06.494	9448-9448	MainActivity	com.example.practical2	D	onResume

PRACTICAL-3

Programs related to different Layouts

Coordinate, Linear, Relative, Table, Absolute, Frame, List View, Grid View.

Layouts in Android Studio control the positioning and arrangement of UI elements on the screen.

- Coordinate Layout: Flexible and dynamic, often used with advanced UI behaviors.
- Linear Layout: Simple, linear arrangement (either vertical or horizontal).
- Relative Layout: Positions views relative to others or the parent container.
- Table Layout: Organizes views in rows and columns, like a table.
- Absolute Layout (deprecated): Positions views at fixed pixel locations (no longer recommended).
- Frame Layout: Stacks views on top of each other.
- List View: Displays scrollable lists of data.
- Grid View: Displays items in a grid format.

3A) Linear Layout

MainActivity.java

```
package com.example.practical3;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:gravity="center"
    android:padding="24dp">

    <!-- Image on Top -->
    <ImageView
        android:layout_width="120dp"
```

```
        android:layout_height="120dp"
        android:src="@drawable/login_icon"
        android:contentDescription="Login Image"
        android:layout_marginBottom="16dp" />

<!-- Title -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Login Page"
    android:textSize="24sp"
    android:textStyle="bold"
    android:layout_marginBottom="24dp" />

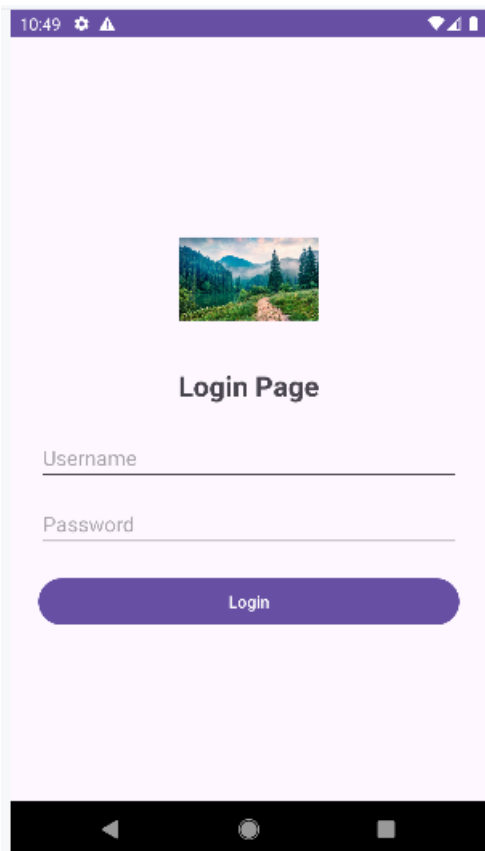
<!-- Username -->
<EditText
    android:id="@+id/etUsername"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Username"
    android:inputType="textPersonName" />

<!-- Password -->
<EditText
    android:id="@+id/etPassword"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:hint="Password"
    android:inputType="textPassword"
    android:layout_marginTop="12dp" />

<!-- Login Button -->
<Button
    android:id="@+id/btnLogin"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:text="Login"
    android:layout_marginTop="20dp" />

</LinearLayout>
```

Output:



3B) Relative Layout

MainActivity.java

```
package com.example.practical3;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
```

```
android:layout_width="match_parent"
android:layout_height="60dp"
android:background="#6200EE"
android:padding="10dp">

<!-- Home -->
<TextView
    android:id="@+id/tvHome"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Home"
    android:textColor="@android:color/white"
    android:textSize="18sp"
    android:layout_alignParentStart="true"
    android:layout_centerVertical="true" />

<!-- About -->
<TextView
    android:id="@+id/tvAbout"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="About"
    android:textColor="@android:color/white"
    android:textSize="18sp"
    android:layout_toEndOf="@id/tvHome"
    android:layout_marginStart="40dp"
    android:layout_centerVertical="true" />

<!-- Contact -->
<TextView
    android:id="@+id/tvContact"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Contact"
    android:textColor="@android:color/white"
    android:textSize="18sp"
    android:layout_toEndOf="@id/tvAbout"
    android:layout_marginStart="40dp"
    android:layout_centerVertical="true" />

<!-- Login -->
<TextView
    android:id="@+id/tvLogin"
    android:layout_width="wrap_content"
```

```
        android:layout_height="wrap_content"
        android:text="Login"
        android:textColor="@android:color/white"
        android:textSize="18sp"
        android:layout_alignParentEnd="true"
        android:layout_centerVertical="true" />
```

</RelativeLayout>

Output:



3C) TableLayout

MainActivity.java

```
package com.example.practical3;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
```

```
public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Activity_main.xml

```
<TableLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:padding="16dp"
    android:stretchColumns="1">

    <TableRow android:background="#FF9800">

        <TextView
            android:text="Item"
            android:textStyle="bold"
            android:padding="8dp"/>

        <TextView
            android:text="Price"
            android:textStyle="bold"
            android:padding="8dp"/>
    </TableRow>

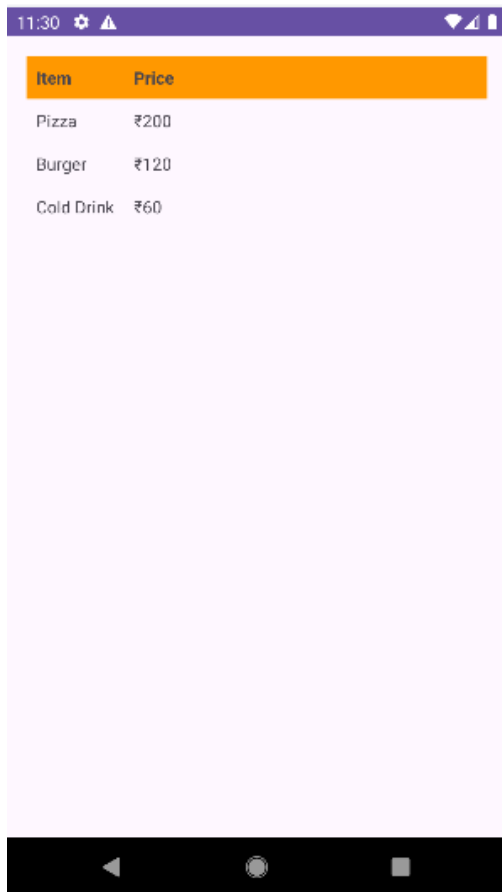
    <TableRow>
        <TextView android:text="Pizza" android:padding="8dp"/>
        <TextView android:text="₹200" android:padding="8dp"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Burger" android:padding="8dp"/>
        <TextView android:text="₹120" android:padding="8dp"/>
    </TableRow>

    <TableRow>
        <TextView android:text="Cold Drink" android:padding="8dp"/>
        <TextView android:text="₹60" android:padding="8dp"/>
    </TableRow>
</TableLayout>
```

</TableLayout>

Output:

A screenshot of an Android application interface. At the top, there is a purple status bar with the time 11:30 and various icons. Below it, the app has a light purple background. A table with an orange header is displayed. The header has two columns: 'Item' and 'Price'. The table contains three rows of data: 'Pizza' with price '₹200', 'Burger' with price '₹120', and 'Cold Drink' with price '₹60'. At the bottom of the screen, there is a black navigation bar with three white icons: a triangle, a circle, and a square.

Item	Price
Pizza	₹200
Burger	₹120
Cold Drink	₹60

3D)Absolute Layout

MainActivity.java

```
package com.example.practical3;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<AbsoluteLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:background="#E3F2FD">

    <!-- Heading -->
    <TextView
        android:layout_width="250dp"
        android:layout_height="wrap_content"
        android:text="Student Registration"
        android:textSize="20sp"
        android:textStyle="bold"
        android:layout_x="60dp"
        android:layout_y="30dp" />

    <!-- Name -->
    <TextView
        android:text="Name"
        android:layout_x="40dp"
        android:layout_y="100dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:hint="Enter Name"
        android:layout_x="140dp"
        android:layout_y="90dp" />

    <!-- Roll Number -->
    <TextView
        android:text="Roll No"
        android:layout_x="40dp"
        android:layout_y="160dp"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />

    <EditText
```

```
        android:layout_width="200dp"
        android:layout_height="wrap_content"
        android:hint="Enter Roll No"
        android:layout_x="140dp"
        android:layout_y="150dp" />
```

```
<!-- Course -->
```

```
<TextView
    android:text="Course"
    android:layout_x="40dp"
    android:layout_y="220dp"
    android:layout_width="wrap_content"
    android:layout_height="wrap_content" />
```

```
<EditText
    android:layout_width="200dp"
    android:layout_height="wrap_content"
    android:hint="Enter Course"
    android:layout_x="140dp"
    android:layout_y="210dp" />
```

```
<!-- Submit Button -->
```

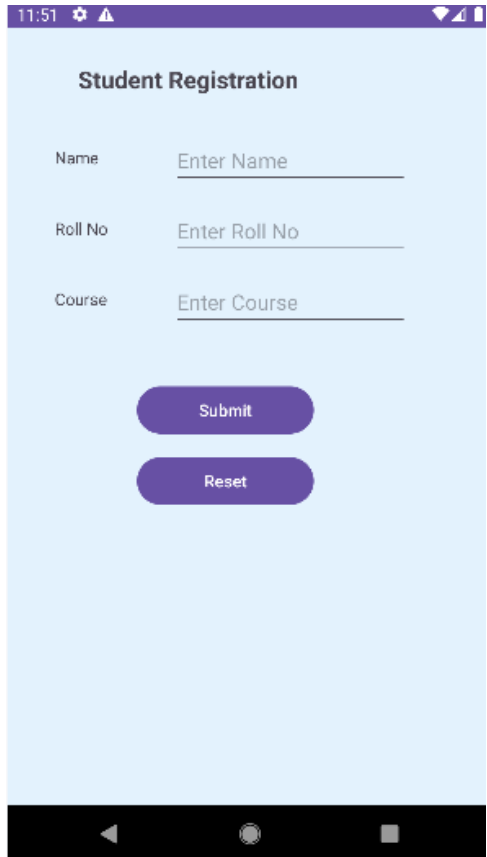
```
<Button
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:text="Submit"
    android:layout_x="110dp"
    android:layout_y="300dp" />
```

```
<!-- Reset Button -->
```

```
<Button
    android:layout_width="150dp"
    android:layout_height="wrap_content"
    android:text="Reset"
    android:layout_x="110dp"
    android:layout_y="360dp" />
```

```
</AbsoluteLayout>
```

Output:



3E)FrameLayout

MainActivity.java

```
package com.example.practical3;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<FrameLayout xmlns:android="http://schemas.android.com/apk/res/android"
```

```
android:layout_width="match_parent"
android:layout_height="match_parent"
android:background="#E0F7FA">

<!-- Profile Image -->
<ImageView
    android:layout_width="150dp"
    android:layout_height="150dp"
    android:src="@android:drawable/sym_def_app_icon"
    android:layout_gravity="top|center_horizontal"
    android:layout_marginTop="60dp" />

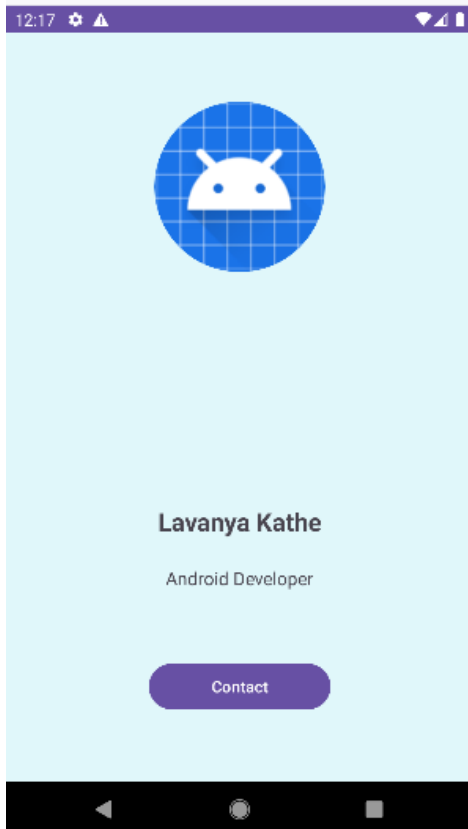
<!-- Name Text -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Lavanya Kathe"
    android:textSize="22sp"
    android:textStyle="bold"
    android:layout_gravity="center"
    android:layout_marginTop="100dp" />

<!-- Course Text -->
<TextView
    android:layout_width="wrap_content"
    android:layout_height="wrap_content"
    android:text="Android Developer"
    android:textSize="16sp"
    android:layout_gravity="center"
    android:layout_marginTop="150dp" />

<!-- Contact Button -->
<Button
    android:layout_width="160dp"
    android:layout_height="wrap_content"
    android:text="Contact"
    android:layout_gravity="bottom|center_horizontal"
    android:layout_marginBottom="60dp" />

</FrameLayout>
```

Output:



3F) ListView Layout

MainActivity.java

```
package com.example.practical3;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.ListView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    ListView listView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        listView = findViewById(R.id.listView);
    }
}
```

```

        String[] subjects = {
            "Mathematics",
            "Science",
            "English",
            "Computer",
            "History",
            "Geography"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            subjects
        );

        listView.setAdapter(adapter);
    }
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

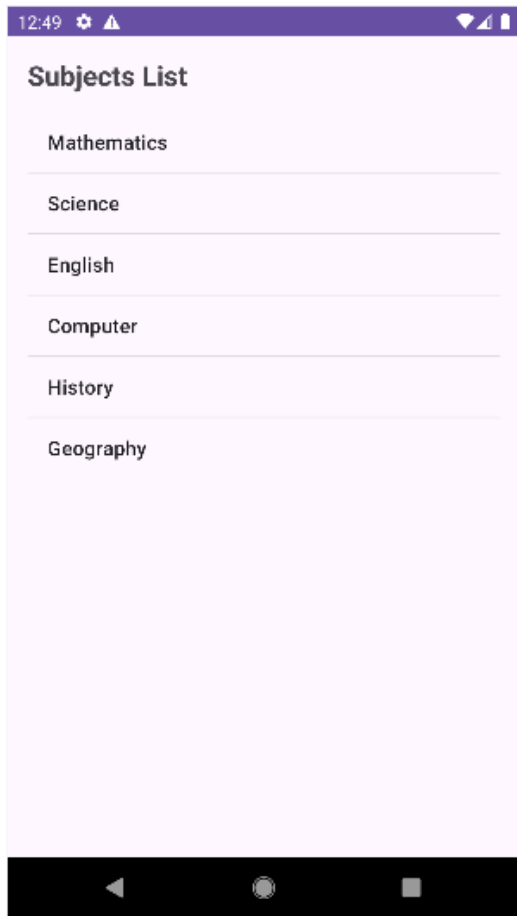
    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Subjects List"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="15dp"/>

    <ListView
        android:id="@+id/listView"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"/>

</LinearLayout>

```

Output:



3G)GridView Layout

MainActivity.java

```
package com.example.practical3;

import android.os.Bundle;
import android.widget.ArrayAdapter;
import android.widget.GridView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    GridView gridView;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
```

```

        setContentView(R.layout.activity_main);

        gridView = findViewById(R.id.gridView);

        String[] subjects = {
            "Maths",
            "Science",
            "English",
            "Computer",
            "History",
            "Geography"
        };

        ArrayAdapter<String> adapter = new ArrayAdapter<>(
            this,
            android.R.layout.simple_list_item_1,
            subjects
        );

        gridView.setAdapter(adapter);
    }
}

```

Activity_main.xml

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Subjects Grid"
        android:textSize="22sp"
        android:textStyle="bold"
        android:layout_marginBottom="15dp"/>

    <GridView
        android:id="@+id/gridView"
        android:layout_width="match_parent"

```

```
android:layout_height="wrap_content"  
android:numColumns="2"  
android:horizontalSpacing="10dp"  
android:verticalSpacing="10dp"/>
```

</LinearLayout>

Output:



PRACTICAL-4

Programming UI elements
AppBar, Fragments, UI Components

a) AppBar

Create a new project, and go to:

ProjectName > App > Src > Main > Res > values > styles.xml Change the default line to this:

(Styles.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <style name="AppTheme" parent="Theme.AppCompat.Light.NoActionBar"></style>
</resources>
```

- Go to, ProjectName > App > Src > Main > Res > Layout
- Right Click on layout and add a new file “toolbar.xml”
- Go to toolbar.xml and change the default layout with this:

(toolbar.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.appcompat.widget.Toolbar
xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:background="@color/design_default_color_secondary"
    android:elevation="4dp">
</androidx.appcompat.widget.Toolbar>
```

MainActivity.java

```
package com.example.practical4;
import android.os.Bundle;
import androidx.appcompat.app.AppCompatActivity;
import androidx.appcompat.widget.Toolbar;

public class MainActivity extends AppCompatActivity {

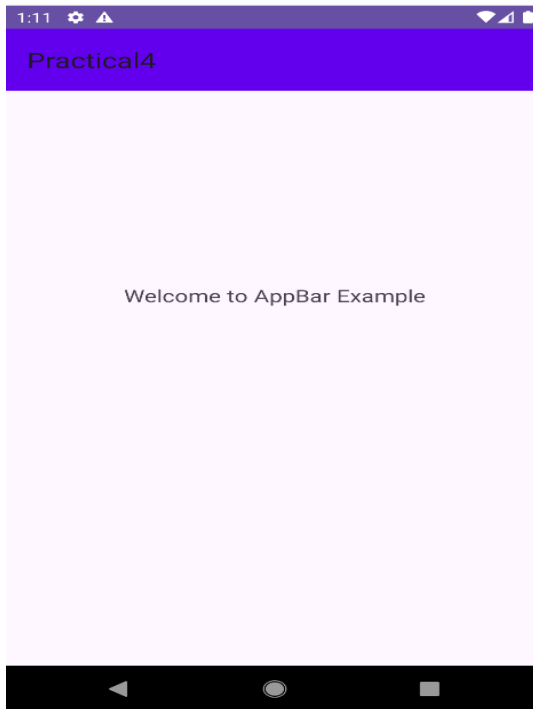
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        Toolbar toolbar = findViewById(R.id.toolbar);
        setSupportActionBar(toolbar);
    }
}
```

```
}  
}
```

Activity_main.xml

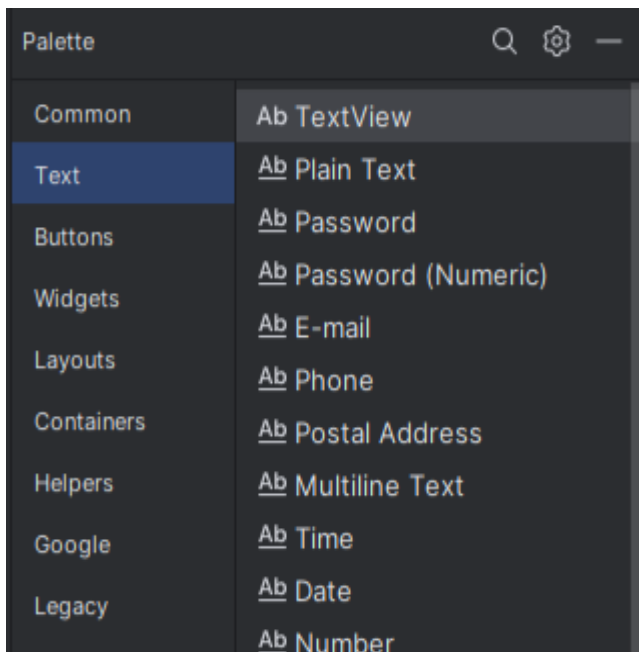
```
<?xml version="1.0" encoding="utf-8"?>  
<LinearLayout  
    xmlns:android="http://schemas.android.com/apk/res/android"  
    xmlns:app="http://schemas.android.com/apk/res-auto"  
    android:layout_width="match_parent"  
    android:layout_height="match_parent"  
    android:orientation="vertical">  
  
    <!-- AppBar / Toolbar -->  
    <androidx.appcompat.widget.Toolbar  
        android:id="@+id/toolbar"  
        android:layout_width="match_parent"  
        android:layout_height="wrap_content"  
        android:background="#6200EE"  
        android:title="My AppBar"  
        android:titleTextColor="#FFFFFF"  
        app:titleCentered="true"/>  
  
    <!-- Page Content -->  
    <TextView  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:text="Welcome to AppBar Example"  
        android:textSize="18sp"  
        android:layout_gravity="center"  
        android:layout_marginTop="200dp"/>  
  
</LinearLayout>
```

Output:



4B)UI Elements

TextView :Label Field In Android, TextView displays text to the user and optionally allows them to edit it programmatically. TextView is a complete text editor, however basic class is configured to not allow editing but we can edit it.



- **EditText:** Input Field In Android, EditText is a standard entry widget in android apps. It is an overlay over TextView that configures itself to be editable. EditText is a subclass of TextView with text editing operations. We often use EditText in our applications in order to provide an input or text field, especially in forms. The most simple example of EditText is Login or Sign-in form.
- **Button:** In Android, Button represents a push button. A Push buttons can be clicked, or pressed by the user to perform an action. There are different types of buttons used in android such as CompoundButton, ToggleButton, RadioButton.

MainActivity.java

```
package com.example.practical4;

import android.os.Bundle;
import android.widget.Button;
import android.widget.TextView;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    private TextView tvDisplay;
    private double firstValue = 0;
    private String operator = "";
    private boolean isNewOp = true;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        tvDisplay = findViewById(R.id.tvDisplay);

        // Numeric Buttons
        int[] numbers = {R.id.btn0, R.id.btn1, R.id.btn2, R.id.btn3,
R.id.btn4, R.id.btn5, R.id.btn6, R.id.btn7, R.id.btn8, R.id.btn9};
        for (int id : numbers) {
            findViewById(id).setOnClickListener(v -> {
                Button b = (Button) v;
                if (isNewOp) tvDisplay.setText(""); // Clear for new
number
                isNewOp = false;
                tvDisplay.append(b.getText().toString());
            });
        }

        // Operator Buttons
```

```

        findViewById(R.id.btnAdd).setOnClickListener(v ->
setOperator("+"));
        findViewById(R.id.btnSub).setOnClickListener(v ->
setOperator("-"));
        findViewById(R.id.btnMul).setOnClickListener(v ->
setOperator("*"));
        findViewById(R.id.btnDiv).setOnClickListener(v ->
setOperator("/"));

        // Clear and Equal
        findViewById(R.id.btnC).setOnClickListener(v -> {
            tvDisplay.setText("0");
            firstValue = 0;
            isNewOp = true;
        });

        findViewById(R.id.btnEqual).setOnClickListener(v ->
calculate());
    }

    private void setOperator(String op) {
        firstValue =
Double.parseDouble(tvDisplay.getText().toString());
        operator = op;
        isNewOp = true;
    }

    private void calculate() {
        double secondValue =
Double.parseDouble(tvDisplay.getText().toString());
        double result = 0;

        switch (operator) {
            case "+": result = firstValue + secondValue; break;
            case "-": result = firstValue - secondValue; break;
            case "*": result = firstValue * secondValue; break;
            case "/":
                if(secondValue != 0) result = firstValue /
secondValue;
                break;
        }
        tvDisplay.setText(String.valueOf(result));
        isNewOp = true;
    }
}

```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="16dp">

    <TextView
        android:id="@+id/tvDisplay"
        android:layout_width="match_parent"
        android:layout_height="100dp"
        android:background="#EEEEEE"
        android:gravity="end|center_vertical"
        android:padding="10dp"
        android:text="0"
        android:textSize="40sp" />

    <LinearLayout android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
        <Button android:id="@+id/btn7" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="7"/>
        <Button android:id="@+id/btn8" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="8"/>
        <Button android:id="@+id/btn9" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="9"/>
        <Button android:id="@+id/btnDiv" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="/" />
    </LinearLayout>

    <LinearLayout android:layout_width="match_parent"
    android:layout_height="wrap_content"
    android:orientation="horizontal">
        <Button android:id="@+id/btn4" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="4"/>
        <Button android:id="@+id/btn5" android:layout_width="0dp"
        android:layout_height="wrap_content" android:layout_weight="1"
        android:text="5"/>
    </LinearLayout>
</LinearLayout>
```

```

        <Button android:id="@+id/btn6" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="6"/>
        <Button android:id="@+id/btnMul" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="*"/>
    </LinearLayout>

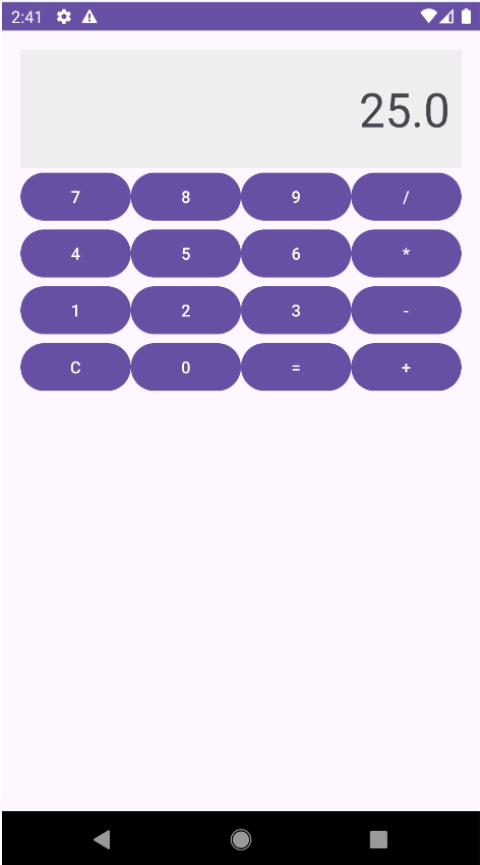
    <LinearLayout android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal">
        <Button android:id="@+id/btn1" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="1"/>
        <Button android:id="@+id/btn2" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="2"/>
        <Button android:id="@+id/btn3" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="3"/>
        <Button android:id="@+id/btnSub" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="-"/>
    </LinearLayout>

    <LinearLayout android:layout_width="match_parent"
android:layout_height="wrap_content"
android:orientation="horizontal">
        <Button android:id="@+id/btnC" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="C"/>
        <Button android:id="@+id/btn0" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="0"/>
        <Button android:id="@+id/btnEqual" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="="/>
        <Button android:id="@+id/btnAdd" android:layout_width="0dp"
android:layout_height="wrap_content" android:layout_weight="1"
android:text="+"/>
    </LinearLayout>

</LinearLayout>

```

Output:



Practical 5

Q] Programming menus, dialog, dialog fragments

Creation of Menu

- Create a new project and go to,
- Project > App -> src -> res(Right-click on res and select) -> New -> Android Resource File -> Filename(menus) and Resource Type Menu
- Now go to, menus.xml file and make the following changes to code file,

(Menu.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<menu xmlns:android="http://schemas.android.com/apk/res/android">
    <item
        android:id="@+id/menu_home"
        android:title="Home" />
    <item
        android:id="@+id/menu_profile"
        android:title="Profile" />
    <item
        android:id="@+id/menu_settings"
        android:title="Settings" />
    <item
        android:id="@+id/menu_help"
        android:title="Help" />
    <item
        android:id="@+id/menu_exit"
        android:title="Exit" />
</menu>
```

Themes.xml file

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <style name="Base.Theme.Prac5aMenus"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <!-- Customize your theme here -->
    </style>
    <style name="Theme.Practical5" parent="Base.Theme.Prac5aMenus"
/>
</resources>
```

MainActivity.java

```
package com.example.prac5amenus;
```

```
import android.os.Bundle;
```

```
import android.view.Menu;
import android.view.MenuItem;
import android.widget.Toast;
import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
    }

    // Create menu
    @Override
    public boolean onCreateOptionsMenu(Menu menu) {
        getMenuInflater().inflate(R.menu.menu_main, menu);
        return true;
    }

    // Handle menu item selection
    @Override
    public boolean onOptionsItemSelected(MenuItem item) {

        int id = item.getItemId();

        if (id == R.id.menu_home) {
            Toast.makeText(this, "Home Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_profile) {
            Toast.makeText(this, "Profile Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_settings) {
            Toast.makeText(this, "Settings Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_help) {
            Toast.makeText(this, "Help Selected", Toast.LENGTH_SHORT).show();
            return true;
        } else if (id == R.id.menu_exit) {
            finish();
            return true;
        }

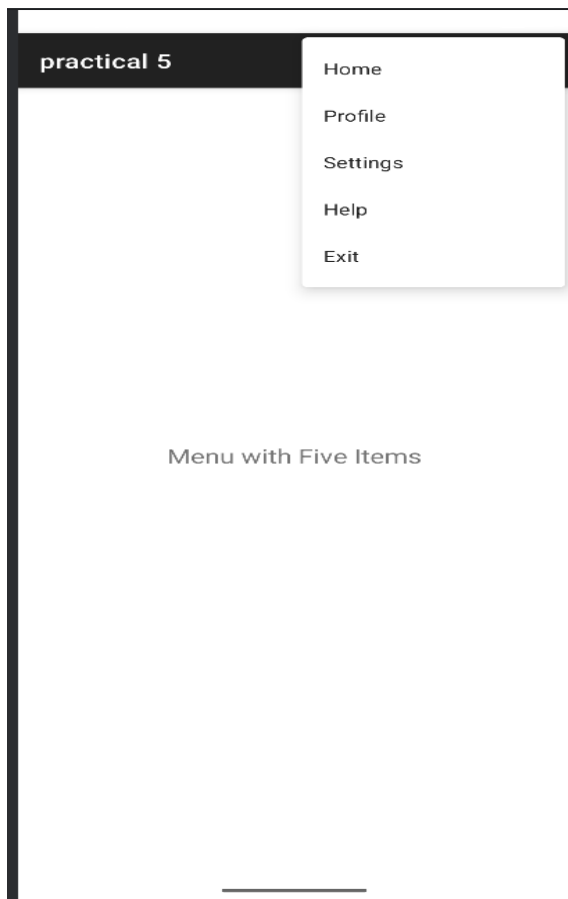
        return super.onOptionsItemSelected(item);
    }
}
```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center"
    android:orientation="vertical">

    <TextView
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Menu with Five Items"
        android:textSize="20sp" />
</LinearLayout>
```

Output :



Creation of Dialog

Dialogs in Android are pop-up windows that prompt the user to take action or provide information. They are used for interactions that require quick feedback or input from the user. There are different types of dialogs in Android, but they can generally be categorized into AlertDialog, ProgressDialog, DatePickerDialog, and

TimePickerDialog. Among them, AlertDialog is the most used.

- Create a new project and
- Add the following code in

(MainActivity.java)

```
package com.example.alertdialogbox; import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity; import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat; import android.content.DialogInterface;
import android.content.Intent;
import androidx.appcompat.app.AlertDialog;
import androidx.appcompat.app.AppCompatActivity; import android.view.View;
import android.widget.Toast; import android.os.Bundle; import android.widget.Button;
public class MainActivity extends AppCompatActivity { Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);
        btn= findViewById(R.id.button);
        btn.setOnClickListener(new View.OnClickListener()
        {
            @Override
            public void onClick(View view) {
                openAlertBox();
            }
        }
    );
}

private void openAlertBox() {
    AlertDialog.Builder builder = new
        AlertDialog.Builder(MainActivity.this);
    builder.setMessage("Do you want to exit ?");
    builder.setIcon(R.drawable.baseline_crisis_ale
rt_24); builder.setTitle("Alert !");
    builder.setCancelable(false);
    builder.setPositiveButton("Yes", (DialogInterface.OnClickListener)
        (dialog, which) -> {
            // When the user click yes button then app
            will close finish();
        });
    builder.setNegativeButton("No",
        (DialogInterface.OnClickListener) (dialog, which) -> {
```

```

        // If user click no then dialog box is canceled.
        dialog.cancel();
    });

    // Create the Alert dialog
    AlertDialog alertDialog = builder.create();
    // Show the Alert Dialog box
    alertDialog.show();
}
}

```

(activity_main.xml)

```

<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto" xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent" android:layout_height="match_parent"
    android:gravity="center" tools:context=".MainActivity">
    <Button
        android:id="@+id/button" android:layout_width="141dp" android:layout_height="79dp"
        android:text="EXIT" />
</LinearLayout>

```

Output :



Practical:-6

Programs on Intents, Events, Listeners and Adapters

The Android Intent Class, Using Events and Event Listeners

In Android, Intents are used for communication between components (e.g., starting activities or services). Events trigger actions (like clicks or touch), while Event Listeners handle these events, executing code in response. Adapters bridge data sources and UI elements (e.g., ListView or RecyclerView), displaying data dynamically.

- Create a new project
- Now we need to create another file, newpage_layout. For this go to, ProjectName > App > src > main > res > layout > New > Layout Resource File > Name the file as newpage_layout > Click OK
- Now we have to create an Empty Activity. For creating an Empty Activity, go to, ProjectName > App > src > main > java > New > Activity > Empty Activity > Name the file as newpage > Finish

MainActivity.java

```
package com.example.prac6;

import android.os.Bundle;

import android.content.Intent;

import android.view.View;

import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnNext;

    @Override
```

```

protected void onCreate(Bundle savedInstanceState) {

    super.onCreate(savedInstanceState);

    setContentView(R.layout.activity_main);

    btnNext = findViewById(R.id.btnNext);

    btnNext.setOnClickListener(new View.OnClickListener() {

        @Override

        public void onClick(View v) {

            Intent intent = new Intent(MainActivity.this, SecondActivity.class);

            startActivity(intent);

        }

    });

}

```

SecondActivity.java

```

package com.example.intentdemo;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override

    protected void onCreate(Bundle savedInstanceState) {

        super.onCreate(savedInstanceState);

        setContentView(R.layout.activity_second);

    }

}

```

Activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical">

    <Button

        android:id="@+id/btnNext"

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Go to Second Activity" />

</LinearLayout>
```

activity_second.xml

```
<?xml version="1.0" encoding="utf-8"?>

<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"

    android:layout_width="match_parent"

    android:layout_height="match_parent"

    android:gravity="center"

    android:orientation="vertical">

    <TextView

        android:layout_width="wrap_content"

        android:layout_height="wrap_content"

        android:text="Welcome to Second Activity"

        android:textSize="20sp"/>
```

```
</LinearLayout>
```

AndroidManifest.xml

```
<?xml version="1.0" encoding="utf-8"?>
```

```
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
```

```
    xmlns:tools="http://schemas.android.com/tools">
```

```
    <application
```

```
        android:allowBackup="true"
```

```
        android:dataExtractionRules="@xml/data_extraction_rules"
```

```
        android:fullBackupContent="@xml/backup_rules"
```

```
        android:icon="@mipmap/ic_launcher"
```

```
        android:label="@string/app_name"
```

```
        android:roundIcon="@mipmap/ic_launcher_round"
```

```
        android:supportsRtl="true"
```

```
        android:theme="@style/Theme.Prac6">
```

```
            <activity
```

```
                android:name=".SecondActivity"
```

```
                android:exported="false"/>
```

```
            <activity android:name=".MainActivity"
```

```
                android:exported="true">
```

```
                <intent-filter>
```

```
                    <action android:name="android.intent.action.MAIN" />
```

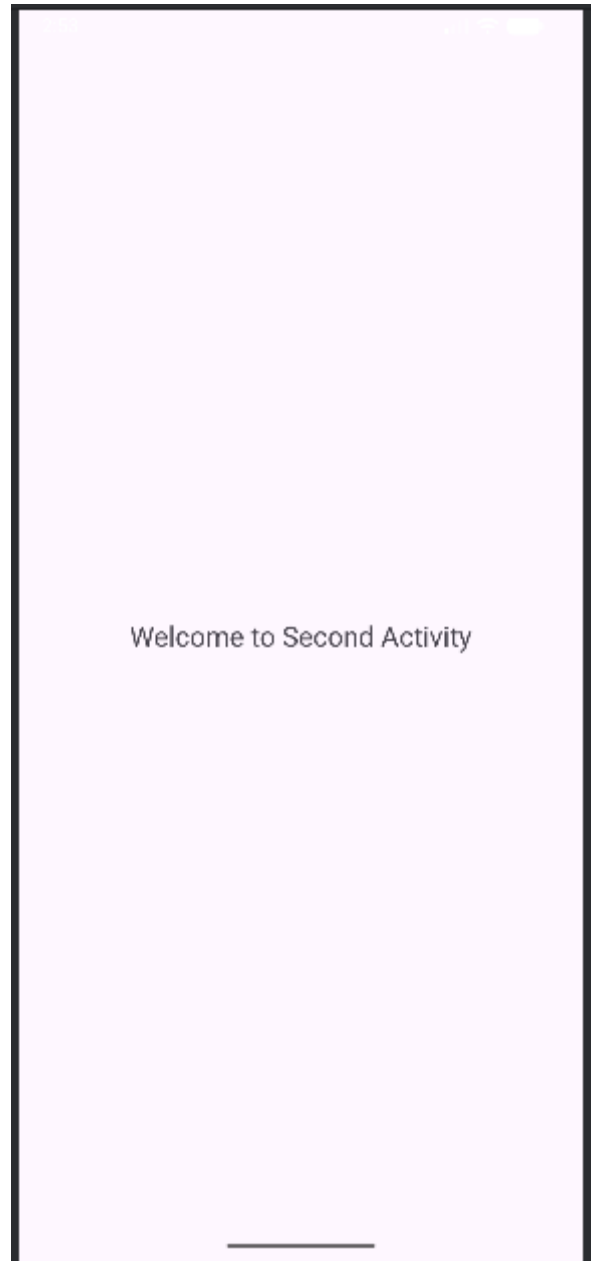
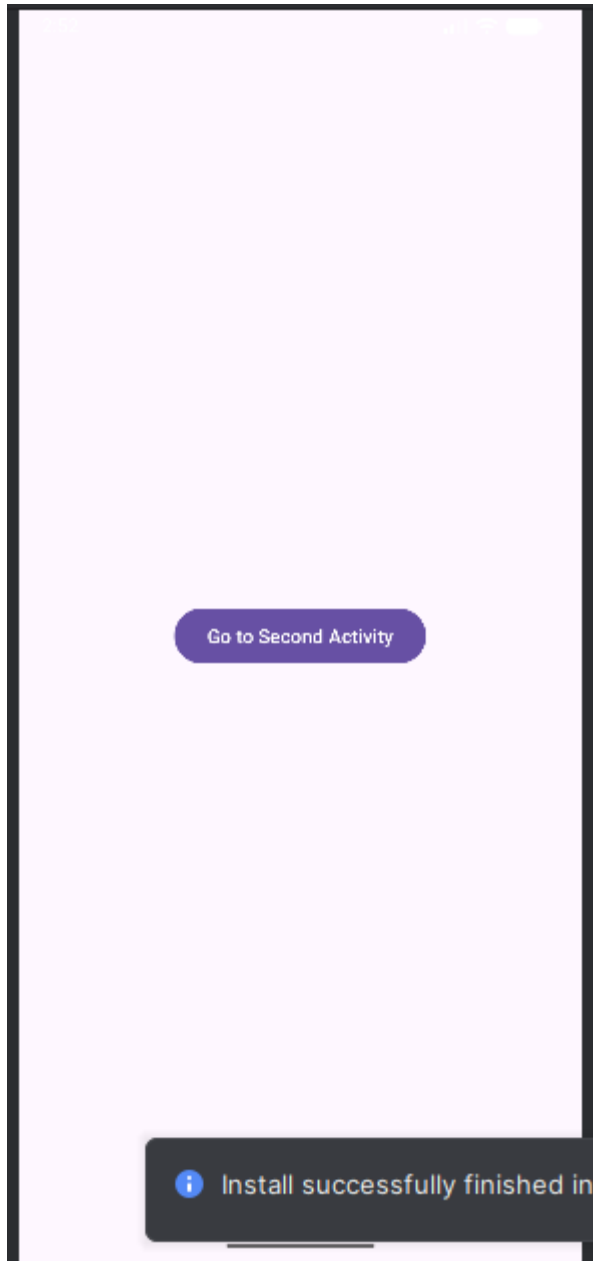
```
                    <category android:name="android.intent.category.LAUNCHER" />
```

```
                </intent-filter>
```

</activity>

</application>

</manifest>



Practical:-7

A. Q] Programs on Services, notification and broadcast receivers

In Android, Services run in the background to perform long-running tasks, such as downloading files or playing music, without a UI. Notifications are used to inform users about important events, even when the app is not in the foreground. Broadcast Receivers listen for system-wide or app specific events, like battery status changes or incoming messages, and respond by triggering specific actions. Together, these components enable apps to function efficiently, interact with users, and handle background processes effectively.

a. Services

- Create a new project
- Right-click on the java package > New > Java Class, name it My Service.
- Declare the Service in AndroidManifest.xml:
- Start the Service

(MainActivity.java)

```
package com.example.prac7a;

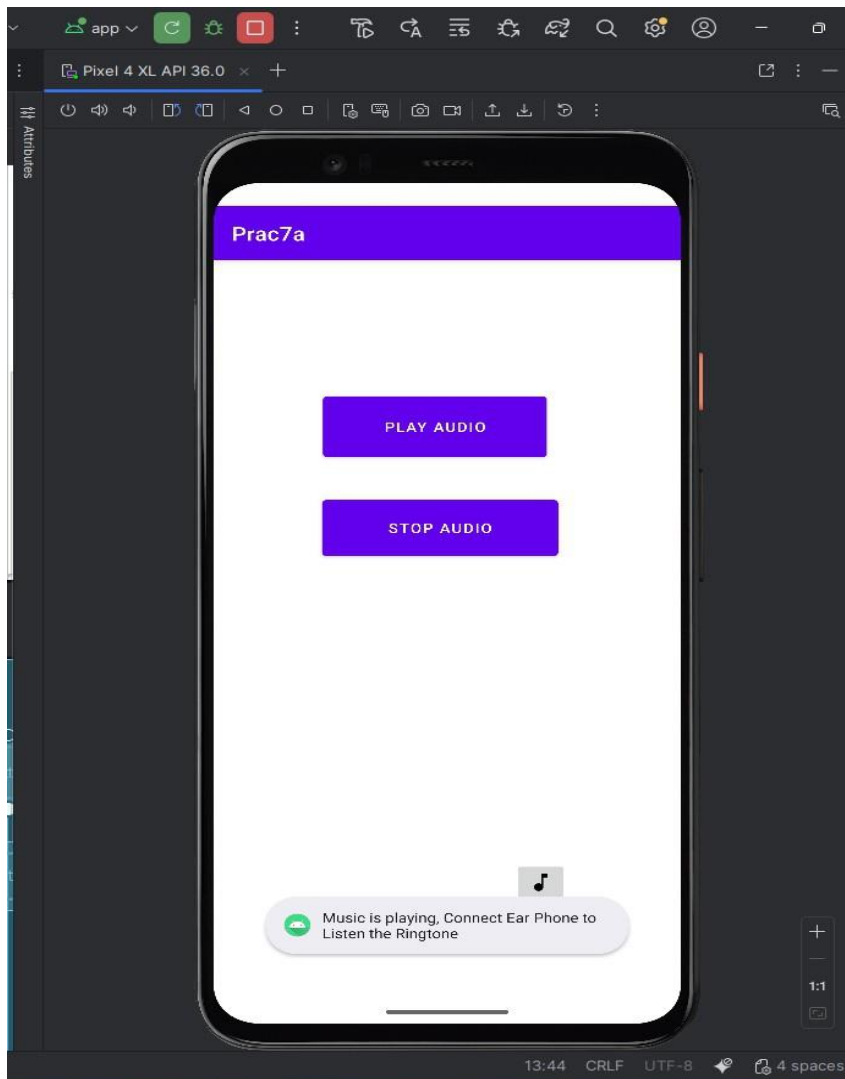
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.os.Bundle;
import android.view.View;
import android.widget.ImageButton;
import android.widget.Toast;
public class MainActivity extends AppCompatActivity { MediaPlayer
mediaPlayer;
    ImageButton ibtn;
    @Override
    protected void onCreate(Bundle savedInstanceState)
{ super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    mediaPlayer = MediaPlayer.create(this,R.raw.myaudio);
}
    public void playAudio(View v)
{
    ibtn = findViewById(R.id.imageButton5);
    ibtn.setVisibility(View.VISIBLE);
    mediaPlayer.start();
    Toast.makeText(this, "Music is playing, Connect Ear Phone to Listen
the Ringtone", Toast.LENGTH_SHORT).show();
}
    public void onPause(View v)
{
    super.onPause(); mediaPlayer.release();
ibtn.setVisibility(View.INVISIBLE);
```

```
}  
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>  
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"  
xmlns:app="http://schemas.android.com/apk/res-auto"  
xmlns:tools="http://schemas.android.com/tools"  
android:layout_width="match_parent" android:layout_height="match_parent"  
tools:context=".MainActivity">  
    <Button  
        android:id="@+id/btnPlay"  
        android:layout_width="197dp"  
        android:layout_height="75dp"  
        android:layout_alignParentEnd="true"  
        android:layout_alignParentRight="true"  
        android:layout_alignParentBottom="true"  
        android:layout_marginEnd="118dp"  
        android:layout_marginRight="118dp"  
        android:layout_marginBottom="579dp"  
        android:onClick="playAudio"  
        android:text="Play Audio"/>  
    <Button  
        android:id="@+id/btnStop"  
        android:layout_width="208dp"  
        android:layout_height="70dp"  
        android:layout_alignParentEnd="true"  
        android:layout_alignParentRight="true"  
        android:layout_alignParentBottom="true"  
        android:layout_marginEnd="108dp"  
        android:layout_marginRight="108dp"  
        android:layout_marginBottom="477dp"  
        android:onClick="onPause"  
        android:text="Stop Audio" />  
    <ImageView  
        android:id="@+id/imageView" android:layout_width="wrap_content"  
        android:layout_height="wrap_content" app:srcCompat="@color/black" />  
    <ImageButton  
        android:id="@+id/imageButton5"  
        android:layout_width="wrap_content"  
        android:layout_height="wrap_content"  
        android:layout_alignParentEnd="true"  
        android:layout_alignParentRight="true"  
        android:layout_alignParentBottom="true"  
        android:layout_marginEnd="99dp"  
        android:layout_marginRight="99dp"  
        android:layout_marginBottom="124dp"  
        android:visibility="invisible"  
        app:srcCompat="@drawable/imagebutton5"  
        tools:visibility="invisible" />  
</RelativeLayout>
```

Output:



B. Notification

(MainActivity.java)

```
package com.example.prac7b;

import android.Manifest;
import android.app.NotificationChannel;
import android.app.NotificationManager;
import android.content.pm.PackageManager;
import android.os.Build;
import android.os.Bundle;
import android.widget.Button;
import android.widget.Toast;

import androidx.annotation.NonNull;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.app.ActivityCompat;
import androidx.core.app.NotificationCompat;
import androidx.core.app.NotificationManagerCompat;
import androidx.core.content.ContextCompat;

public class MainActivity extends AppCompatActivity {

    // Use a NEW channel ID to force Android to create a fresh channel
    private static final String CHANNEL_ID = "default_channel_v2";
    private static final int NOTIFICATION_ID = 1;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        createNotificationChannel();

        Button notifyButton = findViewById(R.id.notifyButton);
        notifyButton.setOnClickListener(v -> checkPermissionAndNotify());
    }

    private void checkPermissionAndNotify() {
        // Android 13+ requires runtime permission for notifications
        if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.TIRAMISU) {
            if (ContextCompat.checkSelfPermission(
                this,
                Manifest.permission.POST_NOTIFICATIONS
            ) != PackageManager.PERMISSION_GRANTED) {

                ActivityCompat.requestPermissions(
                    this,
                    new
String[] {Manifest.permission.POST_NOTIFICATIONS},
                    1
                );
                return;
            }
        }
        sendNotification();
    }

    @Override
    public void onRequestPermissionsResult(
```

```

        int requestCode,
        @NonNull String[] permissions,
        @NonNull int[] grantResults) {

    super.onRequestPermissionsResult(requestCode, permissions,
grantResults);

    if (requestCode == 1) {
        if (grantResults.length > 0
            && grantResults[0] == PackageManager.PERMISSION_GRANTED)
        {
            sendNotification();
        } else {
            Toast.makeText(this,
                "Notification permission denied",
                Toast.LENGTH_SHORT).show();
        }
    }
}

private void sendNotification() {
    NotificationCompat.Builder builder =
        new NotificationCompat.Builder(this, CHANNEL_ID)
            .setSmallIcon(R.mipmap.ic_launcher) // always
provide an icon

            .setContentTitle("应 Notification Test")
            .setContentText("If you see this, notifications
work!")

            .setPriority(NotificationCompat.PRIORITY_HIGH)
            .setDefaults(NotificationCompat.DEFAULT_ALL)
            .setAutoCancel(true);

    NotificationManagerCompat manager =
        NotificationManagerCompat.from(this);

    manager.notify(NOTIFICATION_ID, builder.build());

    Toast.makeText(this, "Notification Sent",
Toast.LENGTH_SHORT).show();
}

private void createNotificationChannel() {
    if (Build.VERSION.SDK_INT >= Build.VERSION_CODES.O) {
        NotificationChannel channel = new NotificationChannel(
            CHANNEL_ID,
            "Default Channel",
            NotificationManager.IMPORTANCE_HIGH
        );
        channel.setDescription("This is the default notification
channel");
        channel.enableVibration(true);
        channel.enableLights(true);

        NotificationManager manager =
            getSystemService(NotificationManager.class);
        manager.createNotificationChannel(channel);
    }
}
}

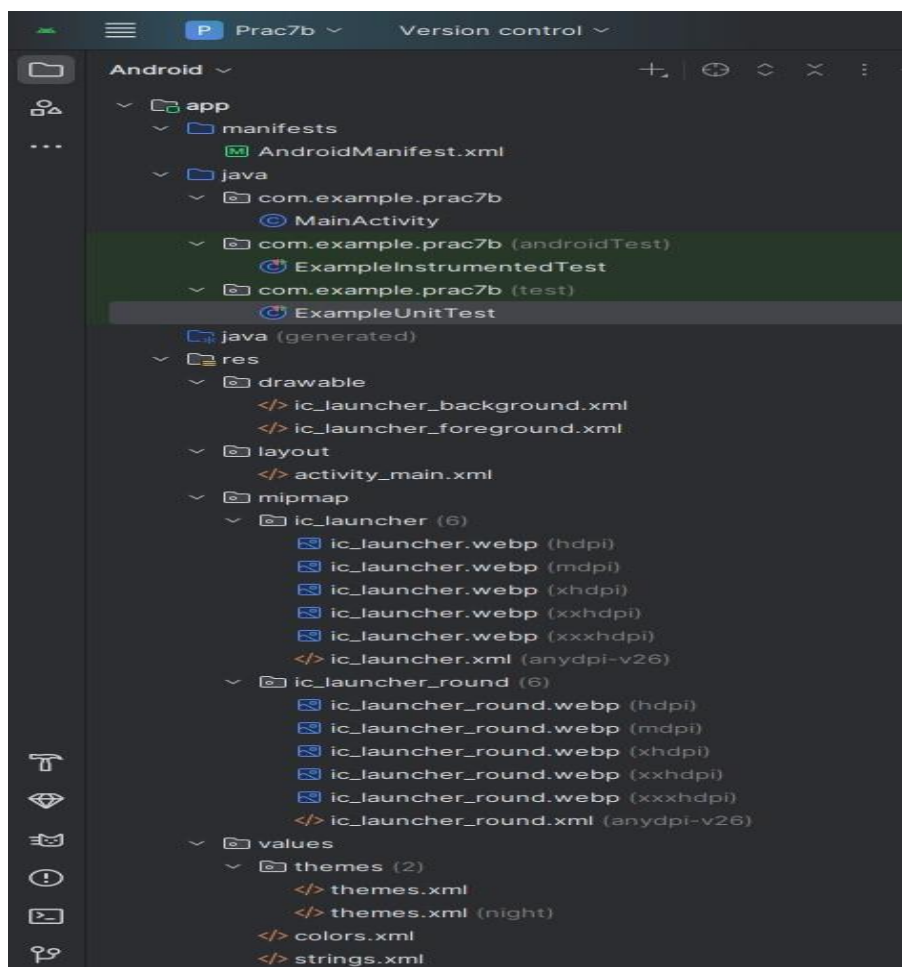
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<androidx.constraintlayout.widget.ConstraintLayout
    xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <Button
        android:id="@+id/notifyButton"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Send Notification"
        app:layout_constraintTop_toTopOf="parent"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintEnd_toEndOf="parent"/>

</androidx.constraintlayout.widget.ConstraintLayout>
```



ExampleInstrumentedTest.java

```
package com.example.prac7b;

import android.content.Context;

import androidx.test.platform.app.InstrumentationRegistry;
import androidx.test.ext.junit.runners.AndroidJUnit4;

import org.junit.Test;
import org.junit.runner.RunWith;

import static org.junit.Assert.*;

/**
 * Instrumented test, which will execute on an Android device.
 *
 * @see <a href="http://d.android.com/tools/testing">Testing
 documentation</a>
 */
@RunWith(AndroidJUnit4.class)
public class ExampleInstrumentedTest {
    @Test
    public void useAppContext() {
        // Context of the app under test.
        Context appContext =
InstrumentationRegistry.getInstrumentation().getTargetContext();
        assertEquals("com.example.prac7b", appContext.getPackageName());
    }
}
```

ic_launcher_background.xml

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path
        android:fillColor="#3DDC84"
        android:pathData="M0,0h108v108h-108z" />
    <path
        android:fillColor="#00000000"
        android:pathData="M9,0L9,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,0L19,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M29,0L29,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M39,0L39,108"
```

```
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M49,0L49,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M59,0L59,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M69,0L69,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M79,0L79,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M89,0L89,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M99,0L99,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,9L108,9"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,19L108,19"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,29L108,29"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,39L108,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,49L108,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,59L108,59"
        android:strokeWidth="0.8"
```

```
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,69L108,69"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,79L108,79"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,89L108,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,99L108,99"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,29L89,29"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,39L89,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,49L89,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,59L89,59"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,69L89,69"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,79L89,79"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M29,19L29,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M39,19L39,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
```

```

<path
    android:fillColor="#00000000"
    android:pathData="M49,19L49,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M59,19L59,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M69,19L69,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
<path
    android:fillColor="#00000000"
    android:pathData="M79,19L79,89"
    android:strokeWidth="0.8"
    android:strokeColor="#33FFFFFF" />
</vector>

```

lc_launcher_foreground.xml

```

<vector xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:aapt="http://schemas.android.com/aapt"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path android:pathData="M31,63.928c0,0 6.4,-11 12.1,-13.1c7.2,-2.6 26,-
1.4 26,-1.4l38.1,38.1L107,108.928l-32,-1L31,63.928z">
        <aapt:attr name="android:fillColor">
            <gradient
                android:endX="85.84757"
                android:endY="92.4963"
                android:startX="42.9492"
                android:startY="49.59793"
                android:type="linear">
                <item
                    android:color="#44000000"
                    android:offset="0.0" />
                <item
                    android:color="#00000000"
                    android:offset="1.0" />
            </gradient>
        </aapt:attr>
    </path>
    <path
        android:fillColor="#FFFFFF"
        android:fillType="nonZero"
        android:pathData="M65.3,45.828l3.8,-6.6c0.2,-0.4 0.1,-0.9 -0.3,-
1.1c-0.4,-0.2 -0.9,-0.1 -1.1,0.3l-3.9,6.7c-6.3,-2.8 -13.4,-2.8 -19.7,0l-
3.9,-6.7c-0.2,-0.4 -0.7,-0.5 -1.1,-0.3C38.8,38.328 38.7,38.828
38.9,39.228l3.8,6.6C36.2,49.428 31.7,56.028 31,63.928h46C76.3,56.028
71.8,49.428 65.3,45.828zM43.4,57.328c-0.8,0 -1.5,-0.5 -1.8,-1.2c-0.3,-0.7 -
0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1
1.2,1.8C45.3,56.528 44.5,57.328 43.4,57.328L43.4,57.328zM64.6,57.328c-0.8,0
-1.5,-0.5 -1.8,-1.2s-0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3
1.2,1 1.2,1.8C66.5,56.528 65.6,57.328 64.6,57.328L64.6,57.328z"

```

```
        android:strokeWidth="1"
        android:strokeColor="#00000000" />
</vector>
```

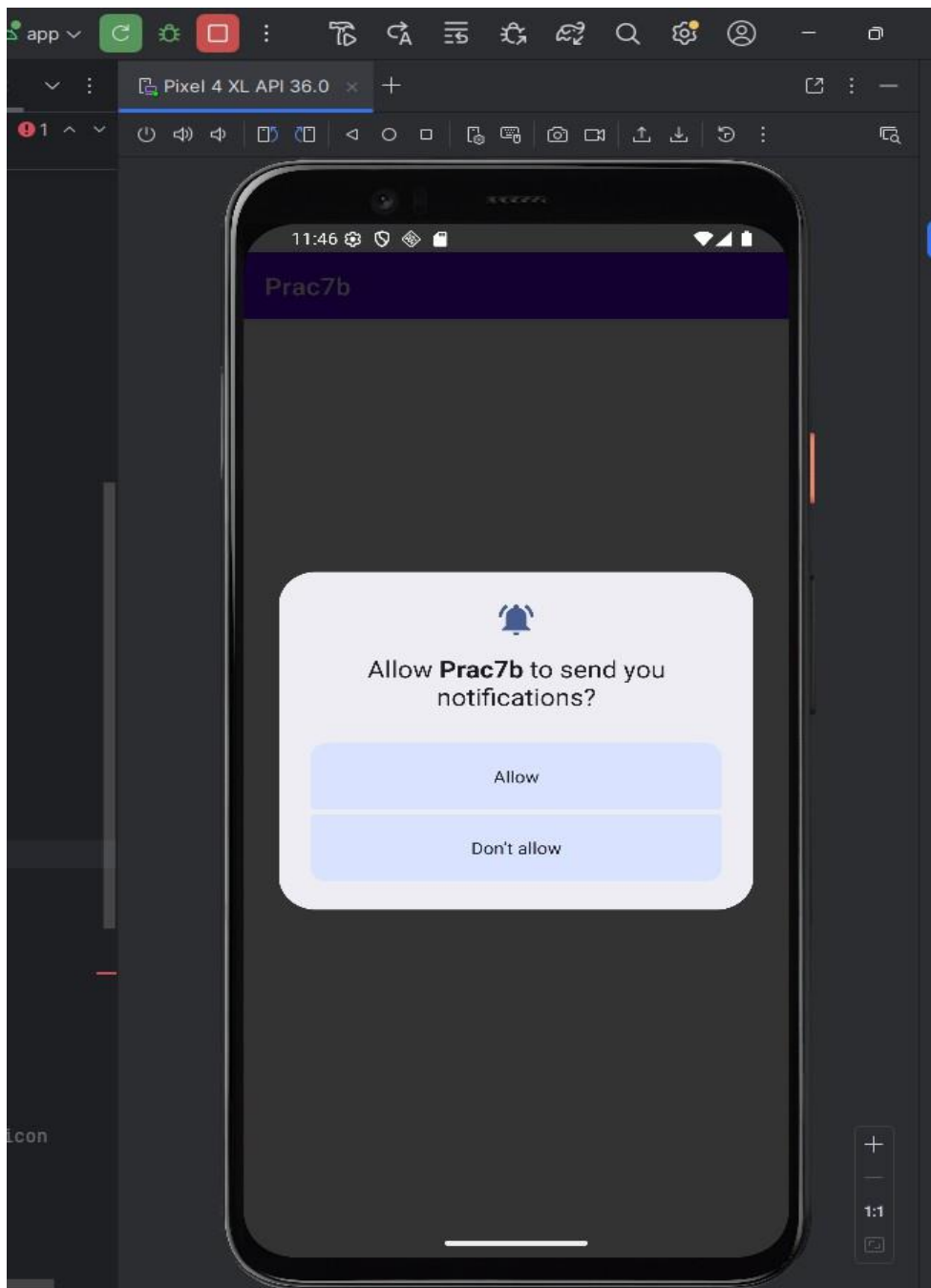
themes.xml

```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Theme.Prac7b"
parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <!-- Primary brand color. -->
        <item name="colorPrimary">@color/purple_500</item>
        <item name="colorPrimaryVariant">@color/purple_700</item>
        <item name="colorOnPrimary">@color/white</item>
        <!-- Secondary brand color. -->
        <item name="colorSecondary">@color/teal_200</item>
        <item name="colorSecondaryVariant">@color/teal_700</item>
        <item name="colorOnSecondary">@color/black</item>
        <!-- Status bar color. -->
        <item
name="android:statusBarColor"?attr/colorPrimaryVariant</item>
        <!-- Customize your theme here. -->
    </style>
</resources>
```

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>
```

Output:



c.broadcast

(MainActivity.java)

```
package com.example.prac7c;

import android.content.Intent;
import android.os.Bundle;
import android.widget.Button;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnNext;

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnNext = findViewById(R.id.btnNext);

        btnNext.setOnClickListener(v -> {
            Intent intent = new Intent(MainActivity.this,
SecondActivity.class);
            startActivity(intent);
        });
    }
}
```

(SecondActivity.java)

```
package com.example.prac7c;

import android.os.Bundle;

import androidx.appcompat.app.AppCompatActivity;

public class SecondActivity extends AppCompatActivity {

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_second);
    }
}
```

(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:gravity="center">

    <Button
        android:id="@+id/btnNext"
```

```
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:text="Go to Second Activity" />

</RelativeLayout>
```

lc_launcher_background.xml

```
<?xml version="1.0" encoding="utf-8"?>
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path
        android:fillColor="#3DDC84"
        android:pathData="M0,0h108v108h-108z" />
    <path
        android:fillColor="#00000000"
        android:pathData="M9,0L9,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,0L19,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M29,0L29,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M39,0L39,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M49,0L49,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M59,0L59,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M69,0L69,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M79,0L79,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M89,0L89,108"
```

```
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M99,0L99,108"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,9L108,9"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,19L108,19"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,29L108,29"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,39L108,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,49L108,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,59L108,59"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,69L108,69"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,79L108,79"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,89L108,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M0,99L108,99"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,29L89,29"
        android:strokeWidth="0.8"
```

```
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,39L89,39"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,49L89,49"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,59L89,59"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,69L89,69"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M19,79L89,79"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M29,19L29,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M39,19L39,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M49,19L49,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M59,19L59,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M69,19L69,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
    <path
        android:fillColor="#00000000"
        android:pathData="M79,19L79,89"
        android:strokeWidth="0.8"
        android:strokeColor="#33FFFFFF" />
</vector>
```

lc_launcher_foreground.xml

```
<vector xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:aapt="http://schemas.android.com/aapt"
    android:width="108dp"
    android:height="108dp"
    android:viewportWidth="108"
    android:viewportHeight="108">
    <path android:pathData="M31,63.928c0,0 6.4,-11 12.1,-13.1c7.2,-2.6 26,-
1.4 26,-1.4l38.1,38.1L107,108.928l-32,-1L31,63.928z">
        <aapt:attr name="android:fillColor">
            <gradient
                android:endX="85.84757"
                android:endY="92.4963"
                android:startX="42.9492"
                android:startY="49.59793"
                android:type="linear">
                <item
                    android:color="#44000000"
                    android:offset="0.0" />
                <item
                    android:color="#00000000"
                    android:offset="1.0" />
            </gradient>
        </aapt:attr>
    </path>
    <path
        android:fillColor="#FFFFFF"
        android:fillType="nonZero"
        android:pathData="M65.3,45.828l3.8,-6.6c0.2,-0.4 0.1,-0.9 -0.3,-
1.1c-0.4,-0.2 -0.9,-0.1 -1.1,0.3l-3.9,6.7c-6.3,-2.8 -13.4,-2.8 -19.7,0l-
3.9,-6.7c-0.2,-0.4 -0.7,-0.5 -1.1,-0.3c38.8,38.328 38.7,38.828
38.9,39.228l3.8,6.6c36.2,49.428 31.7,56.028 31,63.928h46c76.3,56.028
71.8,49.428 65.3,45.828zM43.4,57.328c-0.8,0 -1.5,-0.5 -1.8,-1.2c-0.3,-0.7 -
0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3 1.2,1
1.2,1.8c45.3,56.528 44.5,57.328 43.4,57.328L43.4,57.328zM64.6,57.328c-0.8,0
-1.5,-0.5 -1.8,-1.2s-0.1,-1.5 0.4,-2.1c0.5,-0.5 1.4,-0.7 2.1,-0.4c0.7,0.3
1.2,1 1.2,1.8c66.5,56.528 65.6,57.328 64.6,57.328L64.6,57.328z"
        android:strokeWidth="1"
        android:strokeColor="#00000000" />
</vector>
```

Themes.xml

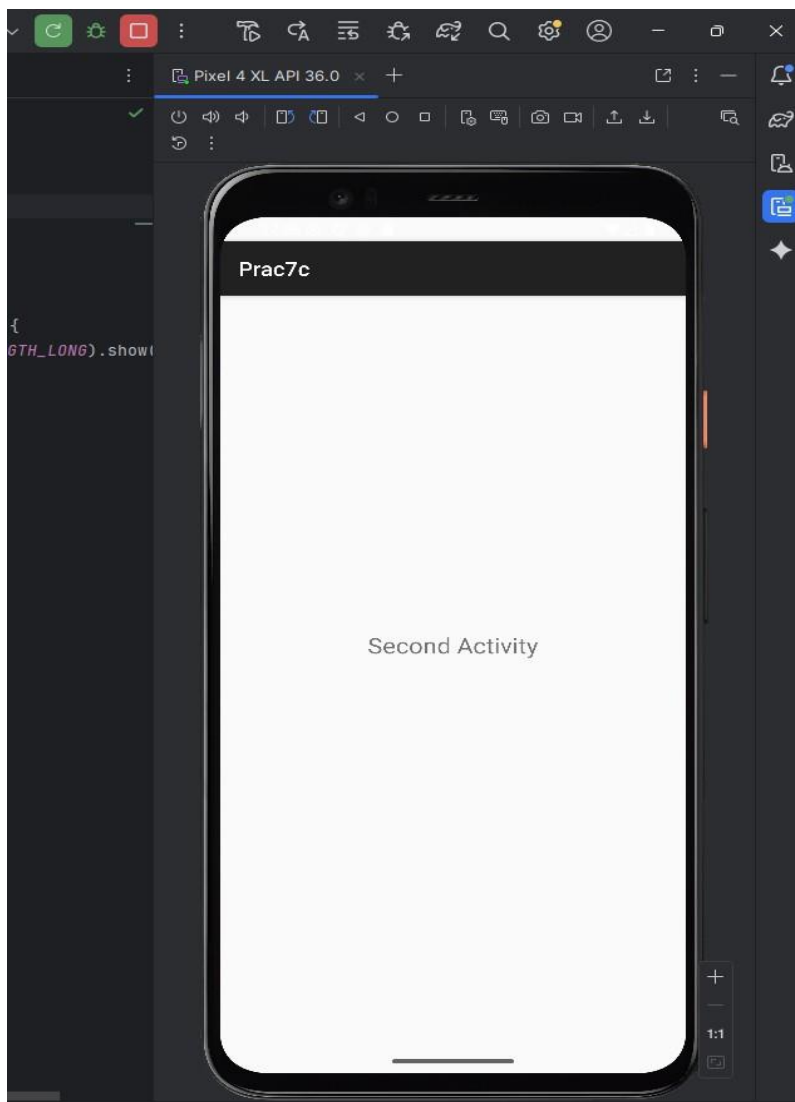
```
<resources xmlns:tools="http://schemas.android.com/tools">
    <!-- Base application theme. -->
    <style name="Theme.Prac7c"
        parent="Theme.MaterialComponents.DayNight.DarkActionBar">
        <!-- Primary brand color. -->
        <item name="colorPrimary">@color/purple_500</item>
        <item name="colorPrimaryVariant">@color/purple_700</item>
        <item name="colorOnPrimary">@color/white</item>
        <!-- Secondary brand color. -->
        <item name="colorSecondary">@color/teal_200</item>
        <item name="colorSecondaryVariant">@color/teal_700</item>
        <item name="colorOnSecondary">@color/black</item>
        <!-- Status bar color. -->
        <item
            name="android:statusBarColor"?attr/colorPrimaryVariant</item>
        <!-- Customize your theme here. -->
```

```
</style>  
</resources>
```

Colors.xml

```
<?xml version="1.0" encoding="utf-8"?>  
<resources>  
    <color name="purple_200">#FFBB86FC</color>  
    <color name="purple_500">#FF6200EE</color>  
    <color name="purple_700">#FF3700B3</color>  
    <color name="teal_200">#FF03DAC5</color>  
    <color name="teal_700">#FF018786</color>  
    <color name="black">#FF000000</color>  
    <color name="white">#FFFFFFFF</color>  
</resources>
```

Output:



Practical:-8

Database programming with SQLite in Android involves using SQLite to store and manage structured data locally within an app. It includes creating a database, defining tables, performing CRUD operations (Create, Read, Update, Delete), and using SQLiteOpenHelper to handle database creation and version management for efficient data storage and retrieval.

- Create a new project
- Right-click your java package > New > Java Class

activity_main.xml

```
<?xml version="1.0" encoding="utf-8"?>

<androidx.constraintlayout.widget.ConstraintLayout
xmlns:android="http://schemas.android.com/apk/res/android"

    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
    android:id="@+id/main"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    tools:context=".MainActivity">

    <TextView
        android:id="@+id/textView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_marginStart="40dp"
        android:layout_marginTop="56dp"
        android:text="Name"
        android:textSize="19sp"
        app:layout_constraintStart_toStartOf="parent"
        app:layout_constraintTop_toTopOf="parent" />

    <EditText
        android:id="@+id/txt_name"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
```

```
android:layout_marginStart="60dp"
android:layout_marginTop="30dp"
android:ems="10"
android:hint=" Enter Name"
android:inputType="text"
app:layout_constraintStart_toEndOf="@+id/textView"
app:layout_constraintTop_toTopOf="parent" />
```

<TextView

```
android:id="@+id/textView2"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="40dp"
android:layout_marginTop="40dp"
android:text="Salary"
android:textSize="19sp"
app:layout_constraintStart_toStartOf="parent"
app:layout_constraintTop_toBottomOf="@+id/textView" />
```

<EditText

```
android:id="@+id/txt_sal"
android:layout_width="wrap_content"
android:layout_height="wrap_content"
android:layout_marginStart="56dp"
android:layout_marginTop="28dp"
android:ems="10"
android:hint=" Enter Salary"
android:inputType="text"
app:layout_constraintStart_toEndOf="@+id/textView2"
app:layout_constraintTop_toBottomOf="@+id/txt_name" />
```

<Button

```
android:id="@+id/btn_add"
android:layout_width="112dp"
```

```
    android:layout_height="54dp"
    android:layout_marginStart="56dp"
    android:layout_marginTop="52dp"
    android:text="ADD"
    app:layout_constraintEnd_toStartOf="@+id/btn_view"
    app:layout_constraintHorizontal_bias="0.27"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/textView2" />
```

<Button

```
    android:id="@+id/btn_view"
    android:layout_width="111dp"
    android:layout_height="54dp"
    android:layout_marginTop="48dp"
    android:layout_marginEnd="72dp"
    android:text="VIEW"
    app:layout_constraintEnd_toEndOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/txt_sal" />
```

<Button

```
    android:id="@+id/btn_del"
    android:layout_width="105dp"
    android:layout_height="55dp"
    android:layout_marginStart="76dp"
    android:layout_marginTop="24dp"
    android:layout_marginBottom="412dp"
    android:text="DELETE"
    app:layout_constraintBottom_toBottomOf="parent"
    app:layout_constraintStart_toStartOf="parent"
    app:layout_constraintTop_toBottomOf="@+id/btn_add" />
```

<Button

```
    android:id="@+id/btn_update"
    android:layout_width="109dp"
    android:layout_height="54dp"
```

```
        android:layout_marginStart="52dp"
        android:layout_marginTop="28dp"
        android:layout_marginBottom="411dp"
        android:text="UPDATE"
        app:layout_constraintBottom_toBottomOf="parent"
        app:layout_constraintStart_toEndOf="@+id/btn_del"
        app:layout_constraintTop_toBottomOf="@+id/btn_view" />
</androidx.constraintlayout.widget.ConstraintLayout>
```

(Android_manifest.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<manifest xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:tools="http://schemas.android.com/tools">

    <application
        android:allowBackup="true"
        android:dataExtractionRules="@xml/data_extraction_rules"
        android:fullBackupContent="@xml/backup_rules"
        android:icon="@mipmap/ic_launcher"
        android:label="@string/app_name"
        android:roundIcon="@mipmap/ic_launcher_round"
        android:supportsRtl="true"
        android:theme="@style/Theme.SQLite">

        <!-- Main Activity -->
        <activity
            android:name=".MainActivity"
            android:exported="true">

            <!-- Launcher intent -->
            <intent-filter>

                <action android:name="android.intent.action.MAIN" />

                <category android:name="android.intent.category.LAUNCHER" />
```

```
        </intent-filter>
    </activity>
</application>
</manifest>
```

(Column_datatype)

```
package com.example.sqlite;

public class Colomn_Datatype {

    int salary;

    String Name ;

}
```

(MainActivity.java)

```
package com.example.sqlite;

import android.os.Bundle;

import android.view.View;

import android.widget.Button;

import android.widget.EditText;

import android.widget.Toast;

import androidx.activity.EdgeToEdge;

import androidx.appcompat.app.AppCompatActivity;

import androidx.core.graphics.Insets;

import androidx.core.view.ViewCompat;

import androidx.core.view.WindowInsetsCompat;

import java.util.ArrayList;

public class MainActivity extends AppCompatActivity {

    mydb dbHelper; // Database helper

    EditText txt_name, txt_sal; // UI components

    Button btn_add, btn_del, btn_view, btn_update; // UI components
```

@Override

```
protected void onCreate(Bundle savedInstanceState) {  
    super.onCreate(savedInstanceState);  
    EdgeToEdge.enable(this);  
    setContentView(R.layout.activity_main);  
    dbHelper = new mydb(this);  
    txt_name = findViewById(R.id.txt_name);  
    txt_sal = findViewById(R.id.txt_sal);  
    btn_add = findViewById(R.id.btn_add);  
    btn_del = findViewById(R.id.btn_del);  
    btn_view = findViewById(R.id.btn_view);  
    btn_update = findViewById(R.id.btn_update);  
    ViewCompat.setOnApplyWindowInsetsListener(findViewById(R.id.main), (v, insets) -> {  
        Insets systemBars = insets.getInsets(WindowInsetsCompat.Type.systemBars());  
        v.setPadding(systemBars.left, systemBars.top, systemBars.right, systemBars.bottom);  
        return insets;  
    });  
    btn_add.setOnClickListener(  
        new View.OnClickListener() {
```

@Override

```
public void onClick(View v) {  
    String name = txt_name.getText().toString().trim();  
    String sala =txt_sal.getText().toString().trim();  
    int sal = Integer.parseInt(sala);  
    if (name.isEmpty() && sala.isEmpty()){  
        Toast.makeText(MainActivity.this, "Any field cannot be empty", Toast.LENGTH_SHORT).show();  
    }  
    else{  
        dbHelper.insert_val(name,sal);  
        Toast.makeText(MainActivity.this, "Data inserted successfully", Toast.LENGTH_SHORT).show();  
    }  
}    }    );
```

```

btn_view.setOnClickListener(
    new View.OnClickListener() {

        @Override
        public void onClick(View v) {

            String name = txt_name.getText().toString().trim();
            String sala =txt_sal.getText().toString().trim();
            int sal = Integer.parseInt(sala);
            ArrayList<Colomn_Datatype> data = dbHelper.fetchdata();
            if (data.isEmpty()) {
                Toast.makeText(MainActivity.this, "No data found", Toast.LENGTH_SHORT).show();
            } else {
                StringBuilder dataString = new StringBuilder();
                for (Colomn_Datatype record : data) {
                    dataString.append("Name: ").append(record.Name).append(", Salary: ")
                }.append(record.salary).append("\n");
                Toast.makeText(MainActivity.this, dataString.toString(), Toast.LENGTH_LONG).show();
            }
        }
    });

btn_update.setOnClickListener(new View.OnClickListener() {
    @Override
    public void onClick(View v) {

        String name = txt_name.getText().toString().trim(); // Name of the record to update
        String sala = txt_sal.getText().toString().trim(); // New salary value

        if (name.isEmpty() || sala.isEmpty()) {
            Toast.makeText(MainActivity.this, "Name and Salary fields cannot be empty",
                Toast.LENGTH_SHORT).show();
        } else {
            int sal = Integer.parseInt(sala);
            boolean isUpdated = dbHelper.updateData(name,sal);
            if (isUpdated) {
                Toast.makeText(MainActivity.this, "Record updated successfully", Toast.LENGTH_SHORT).show();
            } else {

```

```

        Toast.makeText(MainActivity.this, "Record not found or update failed",
Toast.LENGTH_SHORT).show();

    }

    }    }

});

btn_del.setOnClickListener(new View.OnClickListener() {

    @Override

    public void onClick(View v) {

        // Get the name of the employee to delete

        String name = txt_name.getText().toString().trim();

        // Check if the name field is empty

        if (name.isEmpty()) {

            txt_name.setError("Please enter an employee name");

            return; // Stop further execution

        }

        // Call the delete function by name

        int rowsDeleted = dbHelper.deleteEmployeeByName(name);

        // Show a message based on the result

        if (rowsDeleted > 0) {

            Toast.makeText(MainActivity.this, "Employee deleted successfully", Toast.LENGTH_SHORT).show();

        } else {

            Toast.makeText(MainActivity.this, "Employee not found or delete failed",
Toast.LENGTH_SHORT).show();

        }

    }

});

}

}

```

(Mydb)

```

package com.example.sqlite;

import android.content.ContentValues;

import android.content.Context;

import android.database.Cursor;

```

```

import android.database.sqlite.SQLiteDatabase;

import android.database.sqlite.SQLiteOpenHelper;

import androidx.annotation.Nullable;

import java.util.ArrayList;

public class mydb extends SQLiteOpenHelper {

    private static final int db_version = 1;

    private static final String TABLE_EMPLOYEE = "employee";

    private static final String COLUMN_EMP_ID = "emp_id";

    private static final String COLUMN_EMP_NAME = "empname";

    private static final String COLUMN_SALARY = "salary";

    private static final String CREATE_TABLE_EMPLOYEE = "CREATE TABLE " + TABLE_EMPLOYEE + "("

        + COLUMN_EMP_ID + " INTEGER PRIMARY KEY AUTOINCREMENT,"

        + COLUMN_EMP_NAME + " TEXT NOT NULL,"

        + COLUMN_SALARY + " REAL NOT NULL"

        + ")";

    public mydb(@Nullable Context context) {

        super(context, TABLE_EMPLOYEE, null, db_version);

    }

    @Override

    public void onCreate(SQLiteDatabase db) {

        db.execSQL(CREATE_TABLE_EMPLOYEE);

    }

    @Override

    public void onUpgrade(SQLiteDatabase db, int oldVersion, int newVersion) {

        db.execSQL("DROP TABLE IF EXISTS TABLE_EMPLOYEE");

    }

    public void insert_val(String name, int sal){

        SQLiteDatabase myDB = this.getWritableDatabase();

        ContentValues insert = new ContentValues();

        insert.put(COLUMN_EMP_NAME,name);

        insert.put(COLUMN_SALARY,sal);

        myDB.insert(TABLE_EMPLOYEE , null ,insert);

        myDB.close();
    }
}

```

```

    }

    public ArrayList<Colomn_Datatype> fetchdata(){

        SQLiteDatabase myDB = this.getReadableDatabase();

        ArrayList<Colomn_Datatype> arr = new ArrayList<>();

        Cursor cr = myDB.rawQuery("SELECT * FROM "+TABLE_EMPLOYEE,null);

        while(cr.moveToNext()){

            Colomn_Datatype cd = new Colomn_Datatype();

            cd.Name = cr.getString(1);

            cd.salary = cr.getInt(2);

            arr.add(cd);

        }

        return arr;

    }

    public int deleteEmployeeByName(String name) {

        SQLiteDatabase db = this.getWritableDatabase();

        int rowsDeleted = db.delete(TABLE_EMPLOYEE, COLUMN_EMP_NAME + " = ?", new String[]{name});

        db.close();

        return rowsDeleted;

    }


    public boolean updateData(String name, int newSalary) {

        SQLiteDatabase db = this.getWritableDatabase();

        ContentValues contentValues = new ContentValues();

        contentValues.put(COLUMN_SALARY, newSalary); // New salary value


        // Update the record where the name matches

        int rowsAffected = db.update(TABLE_EMPLOYEE, contentValues, COLUMN_EMP_NAME + " = ?", new
String[]{name});

        //db.update(TABLE_EMPLOYEE, contentValues, COLUMN_EMP_NAME + " = " + name, null);

        // Return true if at least one row was updated

        return rowsAffected > 0;

    }

}

```

Running Devices

R, Pix...

+ □

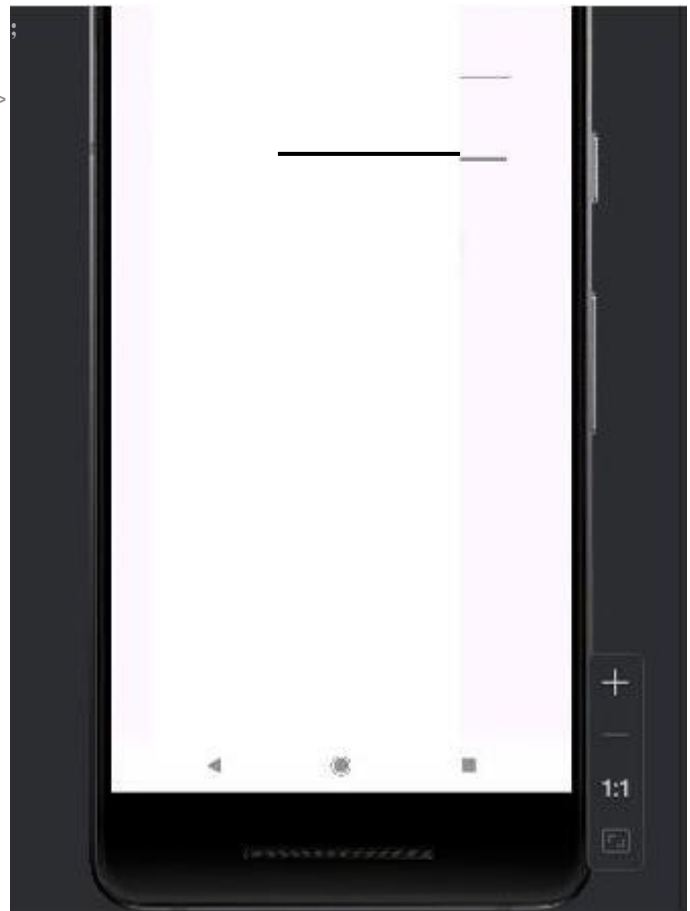
0 4) 4, D 0 <J0 D @ >

Nam•

abc

Salary

12000



Practical:-9

In Android programming, threads run tasks in the background, preventing the UI from freezing. Handlers are used to communicate between threads, like updating the UI from a background thread. Asynchronous programming, often using AsyncTask or Live Data, helps run tasks like network requests without blocking the main UI thread.

- Create a new project
- Add a Text View or Button to interact with or display results
- In MainActivity.java, create a new Thread to perform tasks in the background.
- Click Run in Android Studio to deploy the app to an emulator or physical device

(MainActivity.java)

```
package com.example.practical9b;

import android.os.Bundle;
import android.os.Handler;
import android.view.View;
import android.widget.Button;
import android.widget.TextView;

import androidx.appcompat.app.AppCompatActivity;

public class MainActivity extends AppCompatActivity {

    Button btnStartStop;
    TextView tvTimer;
    int counter = 0;
    boolean isRunning = false;

    Handler handler = new Handler();

    // Runnable that updates the UI every second
    Runnable runnable = new Runnable() {
        @Override
        public void run() {
            counter++;
            tvTimer.setText("Time: " + counter);

            // Repeat every 1 second
            handler.postDelayed(this, 1000);
        }
    };

    @Override
    protected void onCreate(Bundle savedInstanceState) {
        super.onCreate(savedInstanceState);
        setContentView(R.layout.activity_main);

        btnStartStop = findViewById(R.id.btnStartStop);
        tvTimer = findViewById(R.id.tvTimer);

        btnStartStop.setOnClickListener(new View.OnClickListener() {
            @Override
            public void onClick(View v) {
```

```
        if (!isRunning) {
            handler.post(runnable); // Start the timer
            btnStartStop.setText("Stop Timer");
            isRunning = true;
        } else {
            handler.removeCallbacks(runnable); // Stop the timer
            btnStartStop.setText("Start Timer");
            isRunning = false;
        }
    }
});
}
```

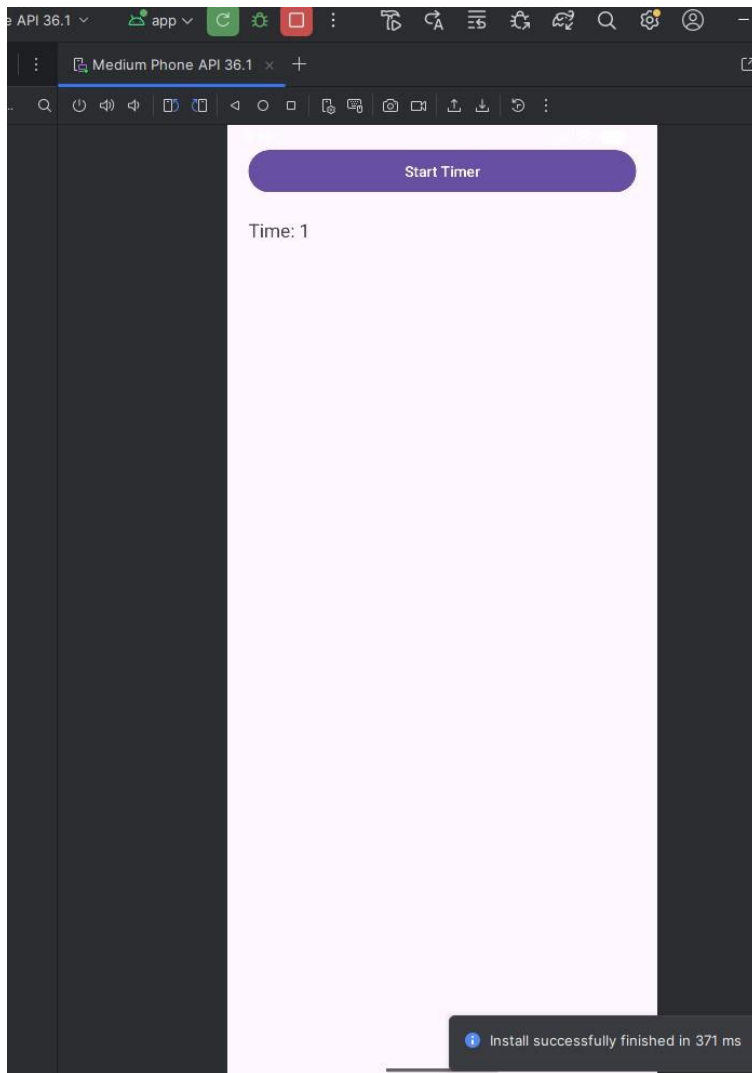
(Activity_main.xml)

```
<?xml version="1.0" encoding="utf-8"?>
<LinearLayout xmlns:android="http://schemas.android.com/apk/res/android"
    android:layout_width="match_parent"
    android:layout_height="match_parent"
    android:orientation="vertical"
    android:padding="20dp">

    <Button
        android:id="@+id/btnStartStop"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Start Timer" />

    <TextView
        android:id="@+id/tvTimer"
        android:layout_width="match_parent"
        android:layout_height="wrap_content"
        android:text="Time: 0"
        android:textSize="18sp"
        android:layout_marginTop="20dp" />
</LinearLayout>
```

Output:



Practical:-10

A] Programming Media API and Telephone API

The Media API in Android allows access to media features like audio, video, and image capture, enabling tasks such as recording, playback, and media file manipulation. The Telephone API provides functionality for managing phone calls, like initiating, receiving, or ending calls, and accessing call-related information.

- Create a new Android project in Android Studio.
- Add the necessary permissions in AndroidManifest.xml
- Use MediaPlayer for playback or MediaRecorder for recording. a. Programming Media API

Code :

A) Media API (Video player)

MainActivity.java :

```
package com.example.prac10a;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.media.MediaPlayer;
import android.net.Uri;
import android.os.Bundle;
import android.os.Environment;
import android.util.Log;
import android.view.View;
import android.widget.Button;
import android.widget.ImageButton; import android.widget.MediaController;
import android.widget.Toast;
import android.widget.VideoView;
public class MainActivity extends AppCompatActivity { VideoView videoview;
    Button btn;
    @Override
    protected void onCreate(Bundle savedInstanceState)
    { super.onCreate(savedInstanceState);
    setContentView(R.layout.activity_main);
    }
    public void playVideo(View view)
    {
        videoview = (VideoView) findViewById(R.id.videoView);

        try {
            MediaController mediacontroller = new
MediaController( MainActivity.this);
            mediacontroller.setAnchorView(videoview);
```

```

        Uri uri = Uri.parse("android.resource://" + getPackageName() +
"/" + R.raw.myvideo);
        videoview.setMediaController(mediacontroller);
videoview.setVideoURI(uri);

    } catch (Exception e) { Log.e("Error", e.getMessage());
e.printStackTrace();
    }
    videoview.requestFocus();
    videoview.setOnPreparedListener(new MediaPlayer.OnPreparedListener()
{

        public void onPrepared(MediaPlayer mp) {

            videoview.start();

        } });
    }
}

```

Activity_main.xml :

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".MainActivity">
    <VideoView
        android:id="@+id/videoView"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" />
    <Button android:id="@+id/btnPlay"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content" android:layout_alignParentEnd="true"
        android:layout_alignParentRight="true"
        android:layout_alignParentBottom="true" android:layout_marginEnd="151dp"
        android:layout_marginRight="151dp" android:layout_marginBottom="274dp"
        android:onClick="playVideo" android:text="Play Video" />
</RelativeLayout>

```

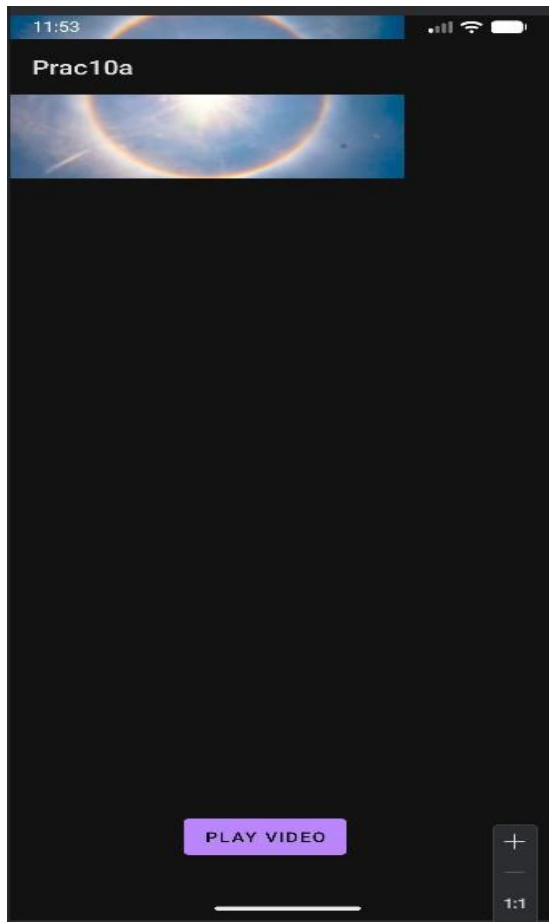
Color.xml :

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>

```

Output :



A.2). Telephone API

- Add the required permissions for phone functionality in AndroidManifest.xml
- Use the TelephonyManager to get phone state and manage calls
- To make a phone call programmatically

A.2) Telephone API:

Code :

MainActivity.java:

```
package com.example.prac10tele;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import android.graphics.Color;
import android.os.Bundle;
import android.app.Activity; import android.content.Intent; import
android.net.Uri; import android.os.Bundle; import android.view.Menu; import
android.view.View; import android.widget.Button;
import android.widget.EditText;
```

```

import android.widget.Toast;
public class MainActivity extends AppCompatActivity { @Override
protected void onCreate(Bundle savedInstanceState)
{ super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    Button dialPhNumber = (Button) findViewById(R.id.dialPhNumber);
    dialPhNumber.setOnClickListener(new View.OnClickListener() {
        public void onClick(View view) {

            Toast.makeText(getApplicationContext(),"Mobile is
Calling",Toast.LENGTH_SHORT).show();
            try {
                EditText phNumber = (EditText) findViewById(R.id.phNumber);

                Intent implicit1 = new Intent(Intent.ACTION_CALL,
Uri.parse("tel:" + phNumber.getText().toString()));
                startActivity(implicit1);
                Toast.makeText(getApplicationContext(),"Mobile is
Calling",Toast.LENGTH_SHORT).show();

            }
            catch (SecurityException e)
            {}
        }
    });
}
}

```

activity_main.xml:

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
xmlns:app="http://schemas.android.com/apk/res-auto"
xmlns:tools="http://schemas.android.com/tools"
android:layout_width="match_parent" android:layout_height="match_parent"
tools:context=".MainActivity">
    <LinearLayout android:layout_width="match_parent"
android:layout_height="match_parent" android:orientation="vertical">
        <EditText
            android:hint="Enter Phone Number" android:id="@+id/phNumber"
            android:layout_width="fill_parent" android:layout_height="81dp"
            android:ems="5" />
        <Button android:id="@+id/dialPhNumber"
            android:layout_width="fill_parent" android:layout_height="45dp"
            android:text="Call Number" />
    </LinearLayout>
</RelativeLayout>

```

Color.xml :

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>

```

Output :

12:02



Prac10tele

9372298902

CALL NUMBER



Mobile is Calling

B] Programming Security and permissions

Programming security and permissions involve controlling access to software resources and data based on user roles and privileges. It ensures that only authorized users can access or modify certain parts of a system. Proper implementation prevents unauthorized actions, safeguarding the integrity and confidentiality of the system.

- Create a New Android Project
- Use Keystore for Secure Data Storage
- Use HTTPS for Network Communication
- Implement Secure Authentication

Code :

(MainActivity.java)

```
package com.example.prac10b;
import android.os.Bundle;
import androidx.activity.EdgeToEdge;
import androidx.appcompat.app.AppCompatActivity;
import androidx.core.graphics.Insets;
import androidx.core.view.ViewCompat;
import androidx.core.view.WindowInsetsCompat;
import androidx.appcompat.app.AppCompatActivity;
import android.graphics.Color;
import android.os.Bundle;
import android.view.View; import android.widget.Button; import
android.widget.EditText; import android.widget.TextView; import
android.widget.Toast;
public class MainActivity extends AppCompatActivity { Button b1,b2;
    EditText ed1,ed2;
    TextView tx1; int counter = 3; @Override
    protected void onCreate(Bundle savedInstanceState)
{ super.onCreate(savedInstanceState);
setContentView(R.layout.activity_main);
    b1 = (Button)findViewById(R.id.button);
    ed1 = (EditText)findViewById(R.id.editText);
    ed2 = (EditText)findViewById(R.id.editText2);
    b2 = (Button)findViewById(R.id.button2);
    tx1 = (TextView)findViewById(R.id.textView3);
    b1.setOnClickListener(new View.OnClickListener() { @Override
    public void onClick(View v) {
        if(ed1.getText().toString().equals("admin") &&
            ed2.getText().toString().equals("admin"))
{ Toast.makeText(getApplicationContext(),
            "Redirecting...",Toast.LENGTH_SHORT).show();
        }else{
            Toast.makeText(getApplicationContext(), "Wrong
Credentials",Toast.LENGTH_SHORT).show();

            tx1.setVisibility(View.VISIBLE);
tx1.setBackgroundColor(Color.RED); counter--;
tx1.setText(Integer.toString(counter));
            if (counter == 0) {
                b1.setEnabled(false);
            }
        }
    }
});
```

```

        b2.setOnClickListener(new View.OnClickListener() { @Override
        public void onClick(View v) { finish();
        }
        });    }}

```

activity_main.xml :

```

<?xml version="1.0" encoding="utf-8"?>
<RelativeLayout xmlns:android="http://schemas.android.com/apk/res/android"
    xmlns:app="http://schemas.android.com/apk/res-auto"
    xmlns:tools="http://schemas.android.com/tools"
        android:layout_width="match_parent"
    android:layout_height="match_parent"
        tools:context=".MainActivity">
    <TextView android:id="@+id/textview"
        android:layout_width="wrap_content"
    android:layout_height="wrap_content" android:layout_alignParentTop="true"
    android:layout_centerHorizontal="true" android:text="Login Example"
    android:textSize="35dp" />

    <EditText
        android:id="@+id/editText"
        android:layout_width="293dp"
        android:layout_height="160dp"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:layout_alignParentEnd="true"
        android:layout_alignParentRight="true"
        android:layout_marginStart="18dp"
        android:layout_marginLeft="18dp"
        android:layout_marginTop="46dp"
        android:layout_marginEnd="100dp"
        android:layout_marginRight="100dp"
        android:focusable="true"
        android:hint="Enter Name"
        android:minHeight="48dp"
        android:padding="16dp"
        android:textColorHighlight="#ff7eff15" />

    <EditText
        android:id="@+id/editText2"
        android:layout_width="match_parent"
        android:layout_height="48dp"
        android:layout_below="@+id/editText"
        android:layout_alignEnd="@+id/editText"
        android:layout_alignRight="@+id/editText"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:layout_marginStart="18dp"
        android:layout_marginLeft="18dp"
        android:layout_marginTop="152dp"
        android:layout_marginEnd="-18dp"
        android:layout_marginRight="-18dp"
        android:ems="10"
        android:hint="Password"
        android:inputType="textPassword"
        android:minHeight="48dp"
        android:padding="16dp" />

    <TextView

```

```

        android:id="@+id/textView2"
        android:layout_width="wrap_content"
        android:layout_height="wrap_content"
        android:layout_below="@+id/editText2"
        android:layout_alignParentStart="true"
        android:layout_alignParentLeft="true"
        android:layout_marginStart="-10dp"
        android:layout_marginLeft="-10dp"
        android:layout_marginTop="96dp"
        android:text="Attempts Remaining"
        android:textSize="25dp" />

        <TextView android:id="@+id/textView3"
            android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:layout_alignTop="@+id/textView2"
            android:layout_alignBottom="@+id/textView2"
            android:layout_alignParentEnd="true"
            android:layout_alignParentRight="true"
            android:layout_toEndOf="@+id/textview"
            android:layout_toRightOf="@+id/textview"
            android:text="Text"
            android:textSize="25dp" />
        <Button android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="login"
            android:id="@+id/button"
            android:layout_alignParentBottom="true"
            android:layout_toLeftOf="@+id/textview"
            android:layout_toStartOf="@+id/textview" />
        <Button android:layout_width="wrap_content"
            android:layout_height="wrap_content"
            android:text="Cancel"
            android:id="@+id/button2"
            android:layout_alignParentBottom="true"
            android:layout_toRightOf="@+id/textview"
            android:layout_toEndOf="@+id/textview" />
    </RelativeLayout>

```

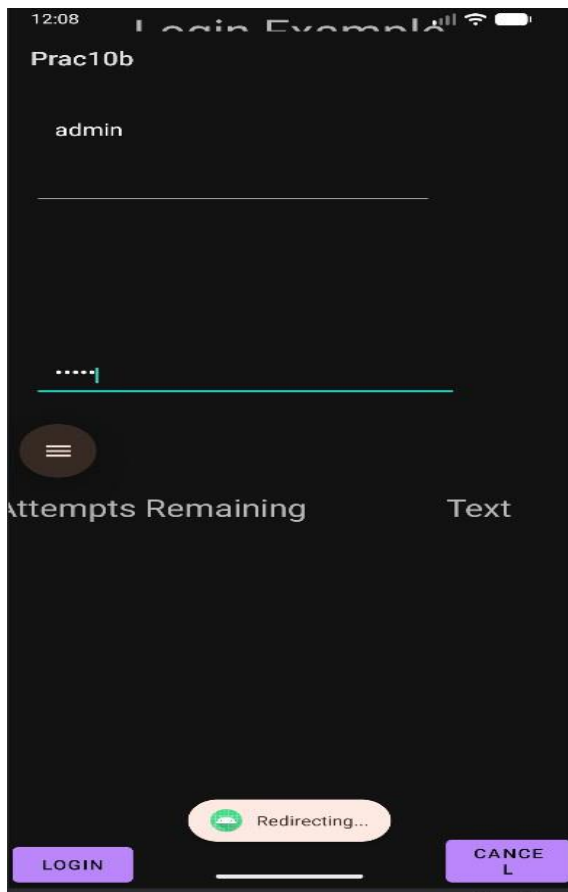
Color.xml :

```

<?xml version="1.0" encoding="utf-8"?>
<resources>
    <color name="purple_200">#FFBB86FC</color>
    <color name="purple_500">#FF6200EE</color>
    <color name="purple_700">#FF3700B3</color>
    <color name="teal_200">#FF03DAC5</color>
    <color name="teal_700">#FF018786</color>
    <color name="black">#FF000000</color>
    <color name="white">#FFFFFFFF</color>
</resources>

```

Output :



For attempt fail

Output :

